GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE

REHABILITATE NORTH RIM UTILITY INFRASTRUCTURE

VOLUME 8 – NORTH RIM WASTEWATER COLLECTION SYSTEM

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES – NORTH RIM
VOLUME 2	PIPELINES – NORTH RIM TO ROARING SPRINGS PUMPHOUSE
VOLUME 3	PIPELINES – NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES – ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE
VOLUME 5	ROARING SPRINGS PUMPHOUSE
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	**NORTH RIM WASTEWATER COLLECTION SYSTEM**
VOLUME 9	POWER DISTRIBUTION – NORTH RIM
VOLUME 10	POWER DISTRIBUTION – NORTH RIM TO ROARING SPRINGS PUMPHOUSE
VOLUME 11	POWER DISTRIBUTION – ROARING SPRINGS PUMPHOUSE TO MANZANITA
VOLUME 12	POWER DISTRIBUTION – COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	080G001	GENERAL	SHEET INDEX
2	080G101	GENERAL	WASTEWATER COLLECTION SYSTEM OVERALL MAP
3	080G102	DEMOLITION	OVERALL TREE REMOVAL
4	081C100	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 1 – A
5	081C101	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 1 – B
6	081C102	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 1 – C
7	081C103	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 2 – A
8	081C104	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 2 – B
9	081C105	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN – REGION 3
–	081C106	DEMOLITION	NOT USED
–	081C107	DEMOLITION	NOT USED
10	082C100	SITEWORK	GESC PLAN – REGION 1 – A
11	082C101	SITEWORK	GESC PLAN – REGION 1 – B
12	082C102	SITEWORK	GESC PLAN – REGION 1 – C
13	082C103	SITEWORK	GESC PLAN – REGION 2 – A
14	082C104	SITEWORK	GESC PLAN – REGION 2 – B
15	082C105	SITEWORK	GESC PLAN – REGION 3
–	082C106	SITEWORK	NOT USED
–	082C107	SITEWORK	NOT USED
–	083C100	LODGE LIFT STATION	NOT USED
–	083C101	LODGE LIFT STATION	NOT USED
–	083C102	LODGE LIFT STATION	NOT USED
–	083C103	LODGE LIFT STATION	NOT USED
–	083C104	LODGE LIFT STATION	NOT USED
–	083C105	LODGE LIFT STATION	NOT USED
–	083C106	LODGE LIFT STATION	NOT USED
–	083C107	LODGE LIFT STATION	NOT USED
–	083E101	LODGE LIFT STATION	NOT USED
–	083E601	LODGE LIFT STATION	NOT USED
–	083Y101	LODGE LIFT STATION	NOT USED
–	083Y601	LODGE LIFT STATION	NOT USED
–	083Y602	LODGE LIFT STATION	NOT USED
–	083Y603	LODGE LIFT STATION	NOT USED

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
16	084C100	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN KEY MAP
17	084C100.1	UTILITIES – ADMIN AREA (REGION 1)	YARD PIPING PLAN AND PROFILE (10" SS)
18	084C101	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINE M
19	084C102	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINES DA, C
20	084C103	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINE A
21	084C104	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINE C – 1
22	084C105	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINE C – 2
23	084C106	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINES D, DE
24	084C107	UTILITIES – ADMIN AREA (REGION 1)	SANITARY SEWER PLAN AND PROFILE – LINES F, AB
25	085C100	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN KEY MAP
26	085C101	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN AND PROFILE – LINES D, M
27	085C102	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN AND PROFILE – LINE A
28	085C103	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN AND PROFILE – LINES E, B, H
29	085C104	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN AND PROFILE – LINES C, F
30	085C105	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN AND PROFILE – LINES G, J
31	086C106	UTILITIES – CAMPGROUND AREA (REGION 2)	SANITARY SEWER PLAN KEY PROFILE – LINES E–A. E–B
32	085C100	UTILITIES – CONCESSIONAIRE AREA (REGION 3)	SANITARY SEWER PLAN KEY MAP
33	086C101	UTILITIES – CONCESSIONAIRE AREA (REGION 3)	SANITARY SEWER PLAN AND PROFILE – LINES C
34	086C102	UTILITIES – CONCESSIONAIRE AREA (REGION 3)	SANITARY SEWER PLAN AND PROFILE – LINES AC, A, D
35	086C103	UTILITIES – CONCESSIONAIRE AREA (REGION 3)	SANITARY SEWER PLAN AND PROFILE – LINES M, B
–	087C100	UTILITIES – LODGE AREA (REGION 4)	NOT USED
–	087C101	UTILITIES – LODGE AREA (REGION 4)	NOT USED
–	087C102	UTILITIES – LODGE AREA (REGION 4)	NOT USED
–	087C103	UTILITIES – LODGE AREA (REGION 4)	NOT USED
–	087C104	UTILITIES – LODGE AREA (REGION 4)	NOT USED
36	088C100	UTILITIES	DETAILS

PMIS – 318719

Registered Professional Engineer (Corp) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 9/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	Mark	Sheet	REVISION	Date	Initial
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202					
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	▽		CONSTRUCTION DOCUMENTS	AUG 2026	

QUALITY DESIGN CERTIFICATION

☐ Prepared in Accordance with Design Development (Title I) _____ Drawing No.
OR
☐ Variance from Design Development (Title I) Approved by Superintendent on _____ Date
OR
☐ Construction Drawing Not Preceded by Design Development (Title I)

_____ Project Manager _____ Date

NATIONAL PARK SERVICE Department of the Interior

CONSTRUCTION DOCUMENTS

UNITED STATES
DEPARTMENT OF THE INTERIOR

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

TITLE OF DRAWING
VOLUME 8 COVER/SHEET INDEX

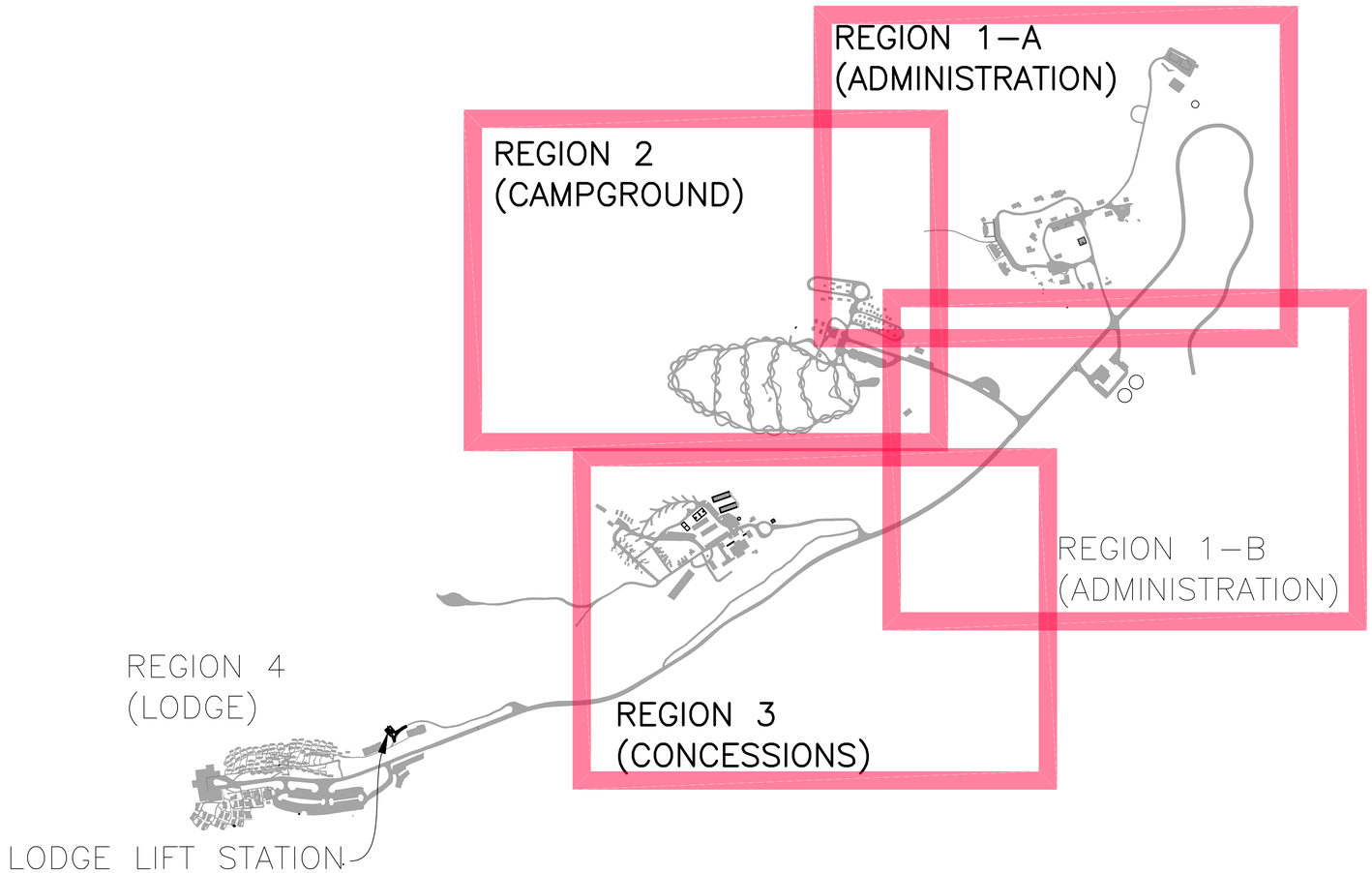
NORTH RIM AND INNERCANYON
NAME OF PARK
GRAND CANYON NATIONAL PARK

REGION	COUNTY	STATE
INTERMOUNTAIN	COCONINO	ARIZONA

DRAWING NO. 113 182752

PKG. NO. 318719

SHEET 1 OF 36



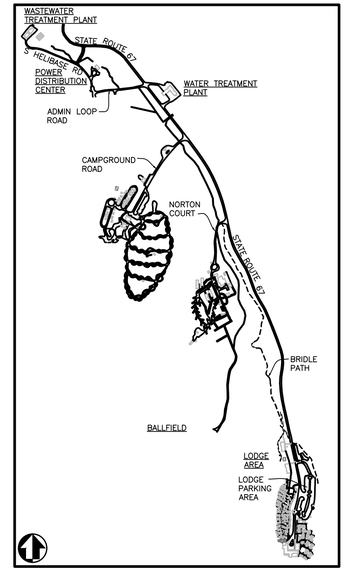
OVERALL MAP

GENERAL NOTES:

1. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW OR REHABILITATED PIPE.
2. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
3. PAVEMENTS TO BE RECONSTRUCTED AT LEAST TO ORIGINAL CONDITION AND PER VOLUME 0 OR SHEET 088C100 DETAILS. CONCRETE PAVEMENT AND CURB AND GUTTER TO BE REPLACED TO NEAREST JOINT.
4. AVAILABLE CONSTRUCTION WIDTH IS INTENDED TO DEFINE THE LIMITS OF CONSTRUCTION CONSIDERING SITE TOPOGRAPHY, INCLUDING TRAIL WIDTH, AND MAXIMUM ALLOWABLE DISTURBANCE.
5. EXISTING SS ALIGNMENT IS APPROXIMATE. CONTRACTOR TO REPLACE SS WITHIN EXISTING ALIGNMENT, UNLESS OTHERWISE NOTED. MINOR DEVIATIONS IN THE NEW PIPELINE ALIGNMENT AND PROFILE ARE ANTICIPATED TO FACILITATE CONSTRUCTION AND TRAIL RESTORATION. REMOVE AND DISPOSE OF DEMOLISHED PIPELINE.
6. IF PROPOSED SS ALIGNMENT DIFFERS FROM EXISTING ALIGNMENT, EXISTING SS PIPE AND STRUCTURES TO BE DEMOLISHED AND REMOVED IN PROXIMITY TO NEW CONSTRUCTION. OTHERWISE EXISTING SS PIPE TO BE ABANDONED FLOWFILL AND CAPPING THE LINE.
7. EXISTING GRADES ARE APPROXIMATE. CONTRACTOR TO MAINTAIN A MINIMUM DEPTH FOR NEW SANITARY LINES OF 3.5 FEET. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2–INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER. CONTRACTOR TO NOTIFY ENGINEER IF COVER OVER PROPOSED PIPE IS LESS THAN 18 INCHES.
8. LENGTHS OF PIPE SHOWN IN PROFILE REFLECTS CALCULATED SLOPE LENGTHS TO BETTER REFLECT TRUE LENGTH OF PIPE TO BE INSTALLED.
9. IT IS CONTRACTOR'S RESPONSIBILITY TO KEEP AT LEAST 3' AWAY FROM EXISTING BUILDINGS AND EXISTING OBJECTS DURING CONSTRUCTION.
10. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND:

EXISTING SANITARY SEWER	
SANITARY SEWER	
EXISTING FORCE MAIN	FM
FORCE MAIN	FM
EXISTING MANHOLES	S
REMOVE AND REPLACE MANHOLES	S
EXISTING UNDERGROUND TELEPHONE	UGT
EXISTING RAW WATER	(RW)
ABANDONED WATER	(*W)
EXISTING POWER LINE	(E/UT)
EXISTING ELECTRIC LINE	(UGE)
EXISTING WATER LINE	(W)
EXISTING NONPOTABLE WATER LINE	(NPW)
EXISTING GAS LINE	(UG)
EXISTING STORM LINE	(SD)
EXISTING DRAINAGE LINE	(D)
EXISTING PAVEMENT	
EXISTING BUILDING FOOTPRINTS	
ASPHALT REPLACEMENT	
GRAVEL ROAD REPLACEMENT	
SIDEWALK REPLACEMENT	
LIMITS OF CONSTRUCTION	
TREE TO BE DEMOLISHED	
AREA OF INTEREST	



KEYMAP

500 0 500 1000

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 080 G101	TITLE OF SHEET GENERAL WASTEWATER COLLECTION SYSTEM OVERALL MAP NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 36

OVERALL TREE DEMOLITION INDEX

REGION 1 TREE DEMOLITION

NUMBER	TYPE	SIZE (IN)
1	PINE	7
2	PINE	9
1	PINE	10
1	PINE	11
1	PINE	12
2	PINE	13
3	PINE	14
2	PINE	15
2	PINE	16
2	PINE	18
1	PINE	19
1	PINE	20
1	PINE	22
1	PINE	23
3	PINE	24
1	PINE	25
3	PINE	26
1	PINE	28
3	PINE	29
5	PINE	30
37	TOTAL PINE TREES	
1	ASPEN	15
3	ASPEN	16
4	TOTAL ASPEN TREES	
86	UNKNOWN	NA
86	TOTAL UNKNOWN TREES	
127	TOTAL TREES REMOVED	

REGION 2 TREE DEMOLITION

NUMBER	TYPE	SIZE (IN)
1	PINE	6
1	PINE	8
1	PINE	10
1	PINE	12
2	PINE	14
2	PINE	16
1	PINE	18
4	PINE	20
2	PINE	24
1	PINE	26
1	PINE	28
8	PINE	30
25	TOTAL PINE TREES	
2	ASPEN	6
2	ASPEN	10
4	TOTAL ASPEN TREES	
94	UNKNOWN	NA
94	TOTAL UNKNOWN TREES	
123	TOTAL TREES REMOVED	

REGION 3 TREE DEMOLITION

NUMBER	TYPE	SIZE (IN)
1	PINE	6
5	PINE	8
2	PINE	10
3	PINE	12
3	PINE	14
3	PINE	16
2	PINE	18
2	PINE	20
7	PINE	22
3	PINE	24
1	PINE	26
1	PINE	28
1	PINE	30
34	TOTAL PINE TREES	
1	ASPEN	7
2	ASPEN	8
1	ASPEN	12
3	ASPEN	14
7	TOTAL ASPEN TREES	
0	UNKNOWN	NA
0	TOTAL UNKNOWN TREES	
41	TOTAL TREES REMOVED	

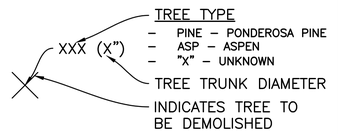
TREE DEMOLITION SUMMARY

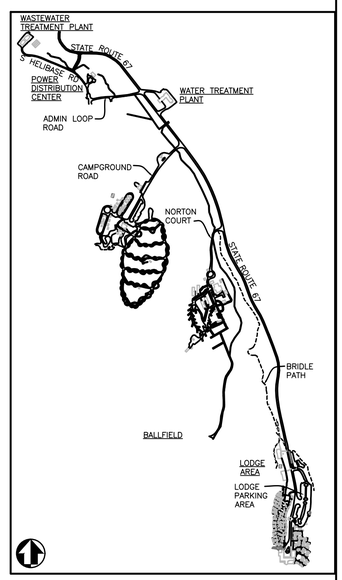
127	REGION 1
118	REGION 2
41	REGION 3
286	TOTAL

GENERAL NOTES:

A. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.
B. REFERENCE SHEET 080G102 FOR TOTAL NUMBER OF TREES FOR THIS REGION TO BE DEMOLISHED PER REGION. PROTECT EXISTING TREES FROM DAMAGE TO THE EXTENT POSSIBLE.
C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.
D. SURVEYED TREES WERE IDENTIFIED, MEASURED, AND LOCATED IN THE FIELD. TREES IDENTIFIED AS "APPROXIMATED TREE" WERE LOCATED VIA AERIAL IMAGERY AND ARE AN ESTIMATED LOCATION OF A LIKELY TREE.
E. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND

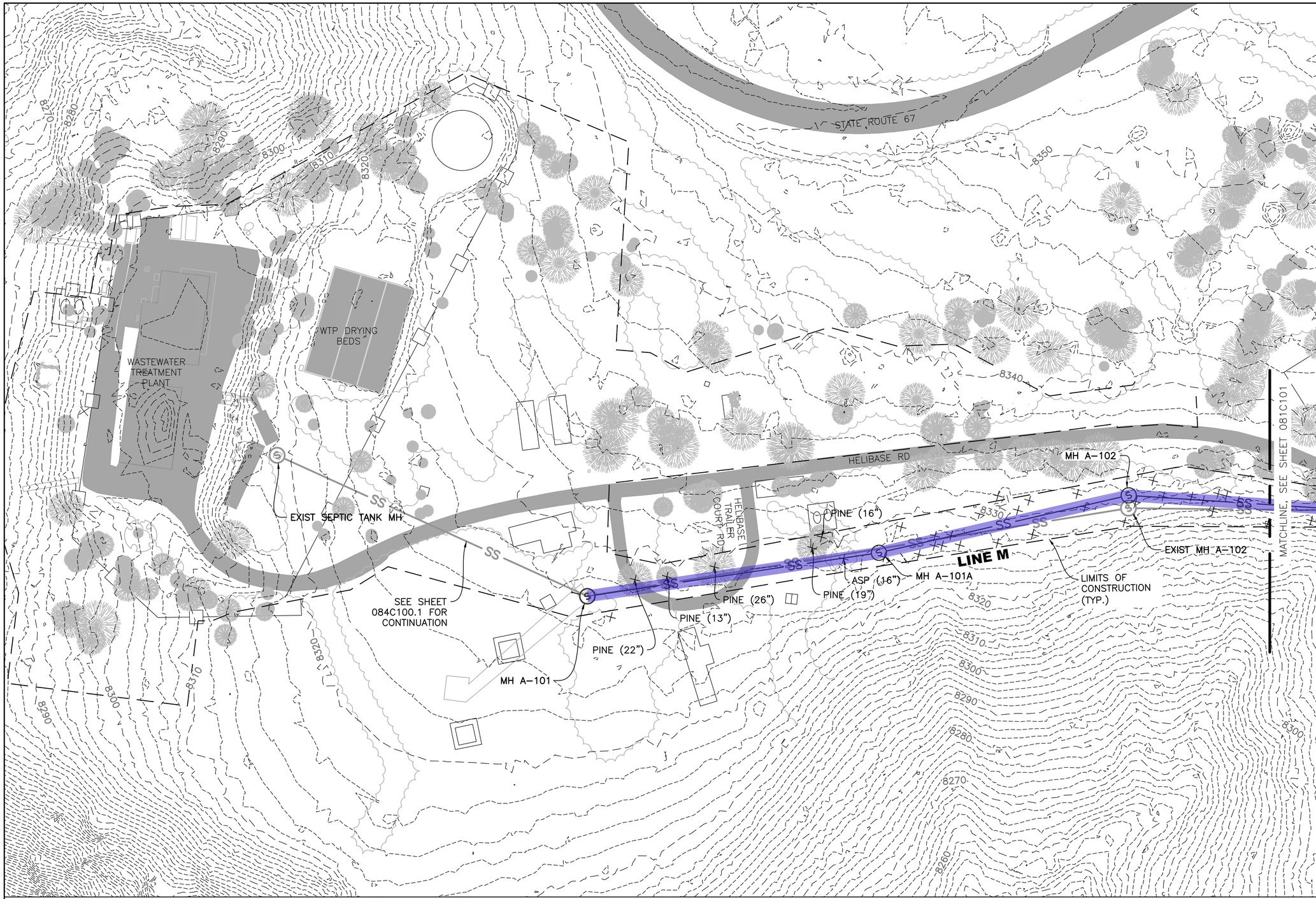




KEYMAP

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

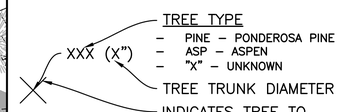
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 080 G102	TITLE OF SHEET DEMOLITION OVERALL TREE REMOVAL NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 36

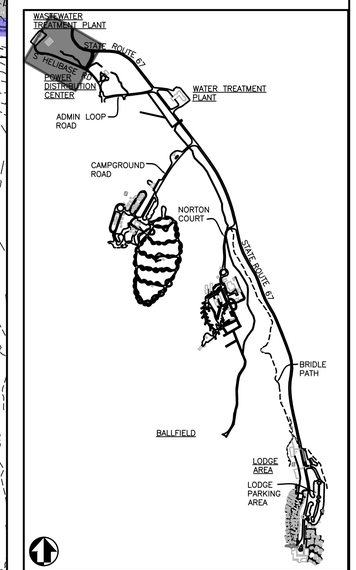


GENERAL NOTES:

A. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.
B. REFERENCE SHEET 080G102 FOR TOTAL NUMBER OF TREES FOR THIS REGION TO BE DEMOLISHED PER REGION. PROTECT EXISTING TREES FROM DAMAGE TO THE EXTENT POSSIBLE.
C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.
D. SURVEYED TREES WERE IDENTIFIED, MEASURED, AND LOCATED IN THE FIELD. TREES IDENTIFIED AS "APPROXIMATED TREE" WERE LOCATED VIA AERIAL IMAGERY AND ARE AN ESTIMATED LOCATION OF A LIKELY TREE.
E. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND





KEYMAP

40 0 40 80
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 5/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	081 C100	DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 1 – A	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DRAFT: SJC		NORTH RIM – GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 4 OF 36
	DATE: AUG 2026			

GENERAL NOTES:

A. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.
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LEGEND

XXX (X") — TREE TYPE
- PINE – PONDEROSA PINE
- ASP – ASPEN
- "X" – UNKNOWN

TREE TRUNK DIAMETER

INDICATES TREE TO BE DEMOLISHED

KEYMAP

40 0 40 80
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 081 C101	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 1 – B NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 36

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

MATCHLINE, SEE SHEET 081C101

MATCHLINE, SEE SHEET 081C103

CLIFFROSE CT

ADMIN SERVICE RD

MH AB-101A

MH D-100

LINE AB

LINE D

EXIST MH AB-100

EXIST MH C-104

MH C-103

LINE C

MH C-102

MH C-101

EXIST MH C-100

EXIST LIFT STATION 3B

LIMITS OF CONSTRUCTION (TYP.)

99.85'

PINE (30")
PINE (25")
PINE (29")

8320
8322
8324
8318
8310
8300
8290
8280
8270
8260
8250
8230
8190

GENERAL NOTES:

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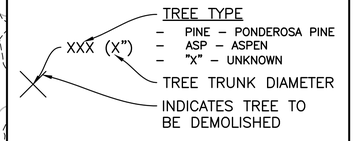
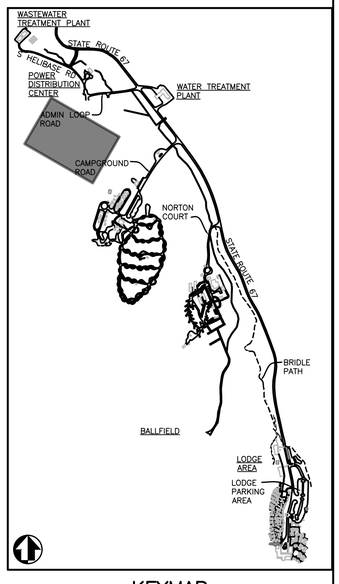
LEGEND





KEYMAP

40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 5/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 081 C102	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 1 – C NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 36

MATCHLINE, SEE SHEET 081C102

MATCHLINE, SEE SHEET 081C104

INN CABIN LOOP

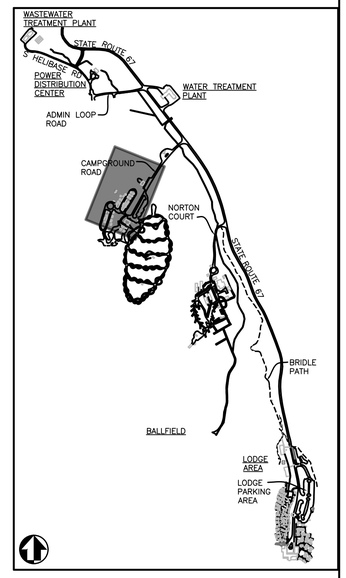
MH EA-201
MH EA-202
LINE E-A
MH EB-200
MH EA-200
MH E-200
LINE E-B
MH B-201A
MH B-201
MH C-200A
MH B-202
MH C-200B
PINE (30")
PINE (30")
TP-15
PINE (28")
PINE (20")
MH C-201A
MH A-201B
LINE C
MH C-201B
PINE (30")
PINE (20")
EXIST MH A-205
PINE (30")
PINE (30")
LINE E
LINE B
MH A-206
PINE (6")
EXIST MH A-201
LIMITS OF CONSTRUCTION (TYP.)
PINE (16")
MH A-207
PINE (24")
MH A-201A
MH A-202A
LINE A
PINE (14")
PINE (30")
MH A-203
EXTERNAL FRAME CABIN LOOP
PINE (20")
PINE (26")
PINE (18")
PINE (10")
EXIST MH A-204A
PINE (16")
MH A-202B
EXIST MH E-201
NORTH RIM GENERAL STORE
CAMPGROUND STORE ACCESS RD
PINE (14")
MH A-206A
ASP (6")
PINE (12")
ASP (10")
PINE (8")
ASP (6")
MH D-200
EXIST MH A-202C
CAMPGROUND RD
ASP (10")
MH D-201
PINE (30")
LINE D
PINE (24")
PINE (20")
EXIST MH D-202
NORTH RIM CAMPGROUND LOOP

GENERAL NOTES:

A. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.
B. REFERENCE SHEET 080G102 FOR TOTAL NUMBER OF TREES FOR THIS REGION TO BE DEMOLISHED PER REGION. PROTECT EXISTING TREES FROM DAMAGE TO THE EXTENT POSSIBLE.
C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.
D. SURVEYED TREES WERE IDENTIFIED, MEASURED, AND LOCATED IN THE FIELD. TREES IDENTIFIED AS "APPROXIMATED TREE" WERE LOCATED VIA AERIAL IMAGERY AND ARE AN ESTIMATED LOCATION OF A LIKELY TREE.
E. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND

XXX (X")
- TREE TYPE
  - PINE – PONDEROSA PINE
  - ASP – ASPEN
  - "X" – UNKNOWN
- TREE TRUNK DIAMETER
- INDICATES TREE TO BE DEMOLISHED



KEYMAP

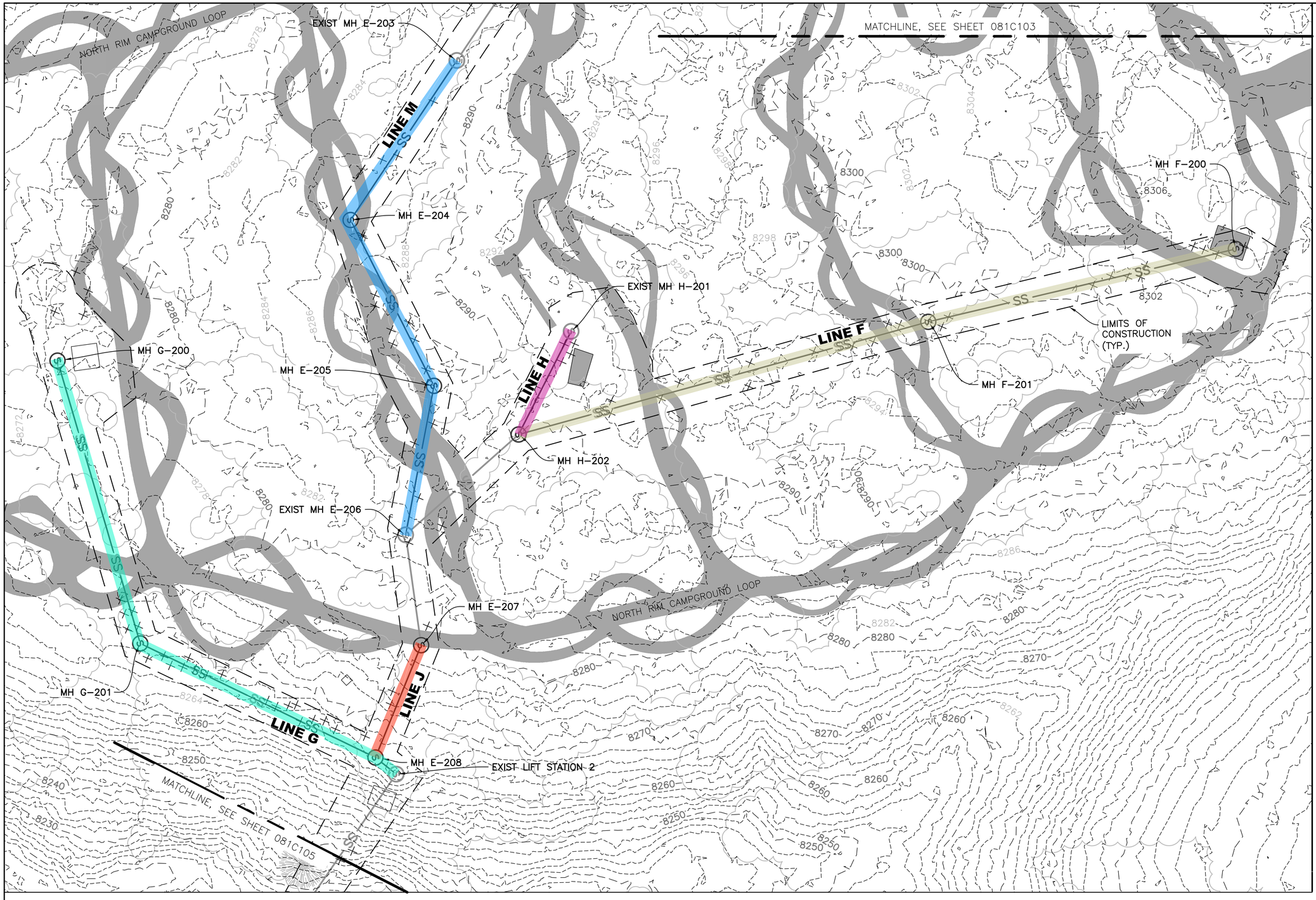
40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	081 C103	DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 2 – A	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DRAWN: SJC		NORTH RIM – GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 7 OF 36
	DATE: AUG 2026			



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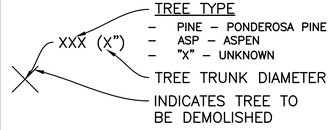
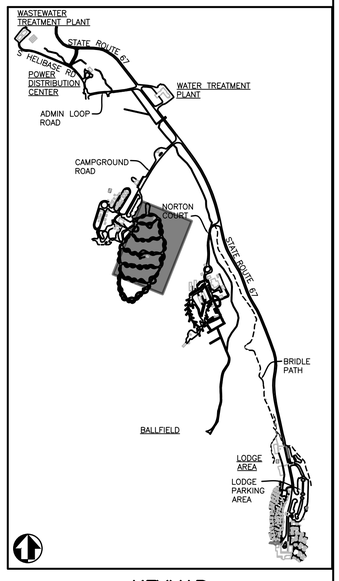
LEGEND





KEYMAP

40 0 40 80
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 5/21/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 081 C104	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 2 – B NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 36

GENERAL NOTES:

A. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.
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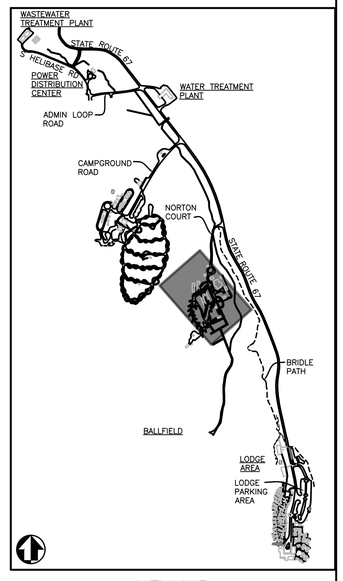
LEGEND

XXX (X") — TREE TYPE
- PINE – PONDEROSA PINE
- ASP – ASPEN
- "X" – UNKNOWN

TREE TRUNK DIAMETER

INDICATES TREE TO BE DEMOLISHED



KEYMAP

40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 6/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 081 C105	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN – REGION 3 NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 36	

GENERAL NOTES:

A. CONSTRUCTION TRUCKS ONLY DRIVE INSIDE OF CONSTRUCTION LIMITS, AND FOLLOW CONSTRUCTION TRUCK PATH ON EXISTING ROAD. SEEDING AND MULCHING SHOULD BE IN DISTURBED AREA INSIDE OF CONSTRUCTION LIMITS.
B. SEE VOLUME 0 FOR EROSION CONTROL DETAILS.
C. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND:

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- SF — SILT FENCE 01/002C510
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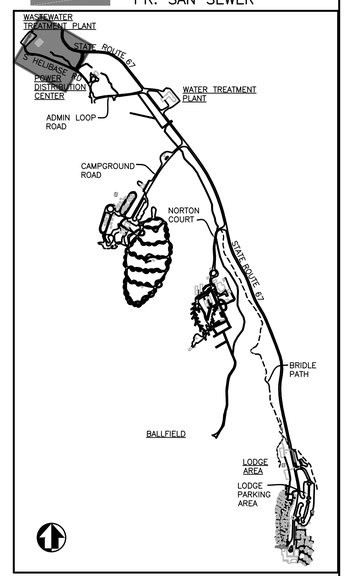
- VTC — VEHICLE TRACKING CONTROL 01/002C511
- CWA — CONCRETE WASH AREA 02/002C511
- LOC — LIMITS OF CONSTRUCTION
- FLOW ARROW
- CONSTRUCTION TRUCK PATH
- PR. SAN SEWER

WASTEWATER TREATMENT PLANT
WTP DRYING BEDS
STATE ROUTE 67
HELIBASE RD
HELIBASE TRAILER COURT RD
MH A-102
N: 13151080.09
E: 1326788.11
LINE M
EXIST MH A-102
N: 13151070.98
E: 1326782.65
LIMITS OF CONSTRUCTION (TYP.)
MATCHLINE, SEE SHEET 082C101
SEE SHEET 084C100.1 FOR CONTINUATION
MH A-101
N: 13151231.25
E: 1326357.47
CONSTRUCTION TURNAROUND LOOP
8260
8270
8280
8290
8300
8310
8320
8330
8340
8350



KEYMAP

40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 082 C100	TITLE OF SHEET SITEWORK GESC PLAN – REGION 1 – A NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 36

MH A–103
N: 13150973.03
E: 1326948.67

MH FM–100
N: 13150825.25
E: 1327119.00

MH FM–100
N: 13150825.25
E: 1327119.00

MH D–102A
N: 13150625.83
E: 1327356.09

MH D–102E
N: 13150476.14
E: 1327499.59

MH D–103 DROP MH
N: 13150519.96
E: 1327665.24

MH D–102
N: 13150371.67
E: 1327543.00

MH D–102D
N: 13150486.85
E: 1327374.39

MH D–102C
N: 13150320.18
E: 1327363.12

MH D–101
N: 13150293.74
E: 1327424.35

MH D–101A
N: 13150138.29
E: 1327397.61

MH A–106
N: 13150256.47
E: 1327329.90

EXIST MH A–104
N: 13150634.88
E: 1326914.67

MH A–104
N: 13150662.58
E: 1326917.45

MH AB–101
N: 13150486.17
E: 1327095.57

MH AB–101B
N: 13150408.31
E: 1327108.40

MH A–106A
N: 13150337.28
E: 1327109.54

MH C–106
N: 13150654.90
E: 1326816.92

MH C–105
N: 13150537.61
E: 1326740.31

MH A–105
N: 13150373.29
E: 1327011.34

LINE F
LINE A
LINE C
LINE DA
LINE DE
LINE DC
LINE AB

CONSTRUCTION TURNAROUND LOOP
CONSTRUCTION TURNAROUND LOOP
CONSTRUCTION TURNAROUND LOOP
CONSTRUCTION TURNAROUND LOOP

NORTH RIM BACKCOUNTRY INFORMATION CENTER
MAINTENANCE BAY
GAS STATION
NATURALIST
PLUMBING SHOP
CARPENTER'S SHOP
LIMITS OF CONSTRUCTION (TYP.)
LIMITS OF CONSTRUCTION (TYP.)

MAINTENANCE LOOP
HELIBASE RD
ADMIN LOOP RD
ADMIN LOOP RD
CLIFFROSE CT

SEE VOLUME 1 FOR CONTINUATION

MATCHLINE, SEE SHEET 082C100
MATCHLINE, SEE SHEET 082C102

GENERAL NOTES:

A. CONSTRUCTION TRUCKS ONLY DRIVE INSIDE OF CONSTRUCTION LIMITS, AND FOLLOW CONSTRUCTION TRUCK PATH ON EXISTING ROAD. SEEDING AND MULCHING SHOULD BE IN DISTURBED AREA INSIDE OF CONSTRUCTION LIMITS.
B. SEE VOLUME 0 FOR EROSION CONTROL DETAILS.
C. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND:

- CF — CONSTRUCTION FENCING 01/002C509
- SF — SILT FENCE 01/002C510
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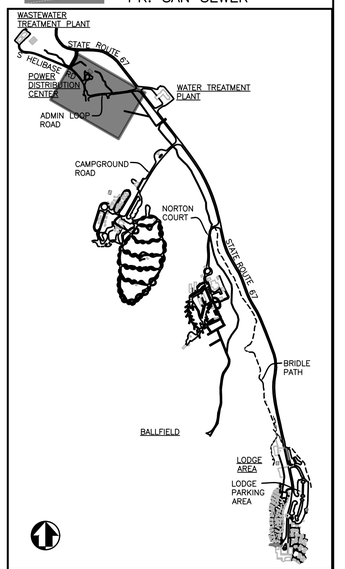
- VTC — VEHICLE TRACKING CONTROL 01/002C511
- CWA — CONCRETE WASH AREA 02/002C511
- LOC — LIMITS OF CONSTRUCTION
- FLOW ARROW
- CONSTRUCTION TRUCK PATH
- PR. SAN SEWER



KEYMAP

40 0 40 80
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 082 C101	TITLE OF SHEET SITEWORK GESC PLAN – REGION 1 – B NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 11 OF 36	

MATCHLINE, SEE SHEET 082C101

MH AB-101A
N: 13150168.75
E: 1327127.78

CLIFFROSE CT

99.85'

MH D-100
N: 13149978.76
E: 1327304.48

LINE AB

EXIST MH C-104
N: 13150211.96
E: 1326792.34

MH C-103
N: 13150119.96
E: 1326832.03

LINE C

EXIST MH AB-100
N: 13150028.95
E: 1327142.70

ADMIN SERVICE RD

LINE D

MH C-102
N: 13150028.77
E: 1326871.69

50' TYP.

EXIST LIFT STATION 3B
N: 13149737.45
E: 1327115.19

MH C-101
N: 13149904.42
E: 1326926.19

LIMITS OF CONSTRUCTION (TYP.)

EXIST MH C-100
N: 13149712.30
E: 1327021.09

MATCHLINE, SEE SHEET 082C103

GENERAL NOTES:

A. CONSTRUCTION TRUCKS ONLY DRIVE INSIDE OF CONSTRUCTION LIMITS, AND FOLLOW CONSTRUCTION TRUCK PATH ON EXISTING ROAD. SEEDING AND MULCHING SHOULD BE IN DISTURBED AREA INSIDE OF CONSTRUCTION LIMITS.
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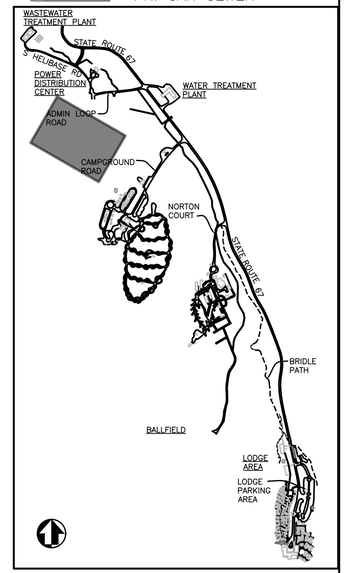

KEYMAP

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SCALE OF FEET

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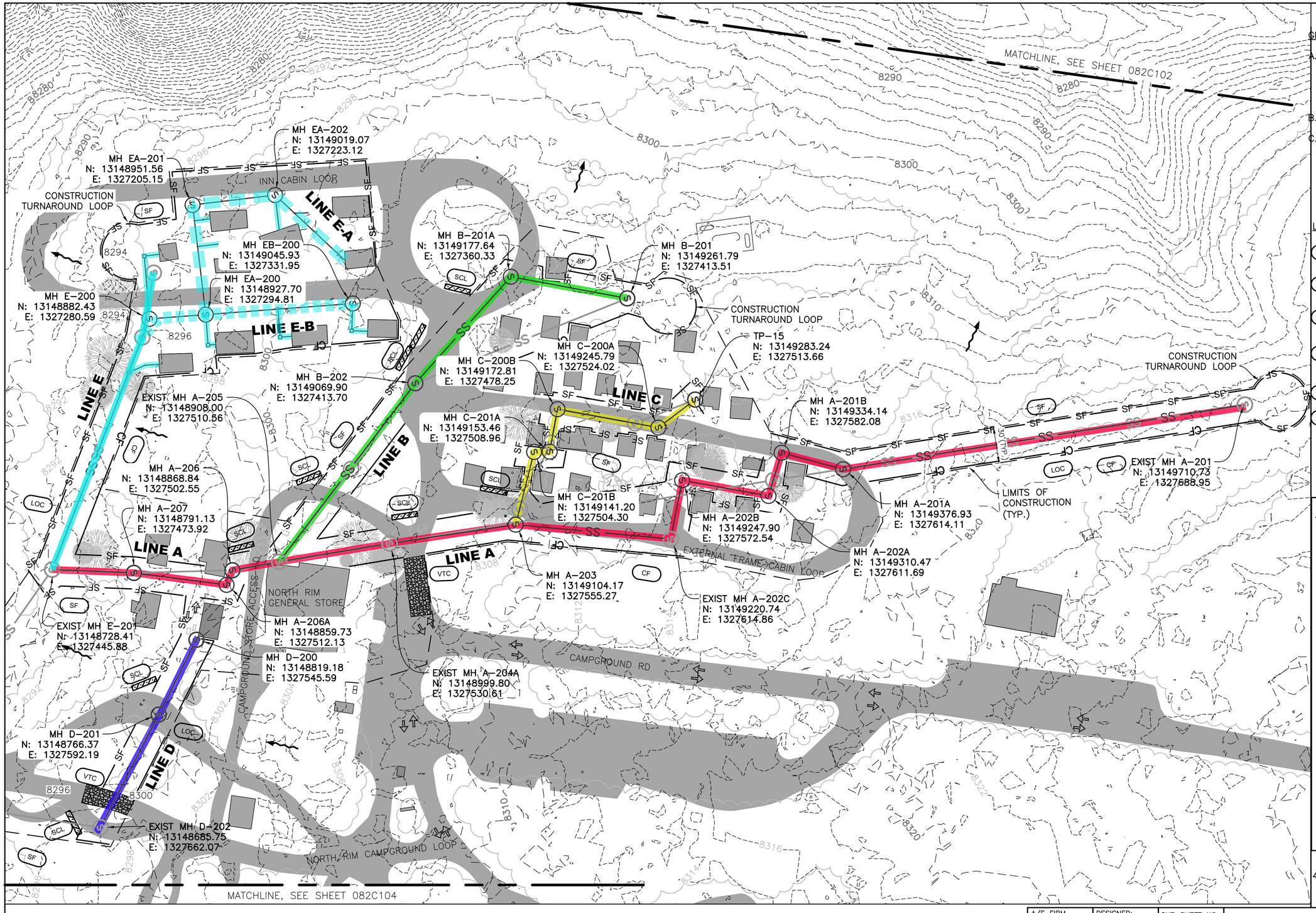

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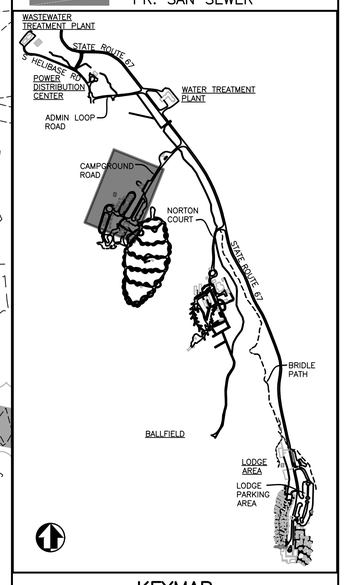

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Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	082 C103	SITEWORK GESC PLAN – REGION 2 – A
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DRAWN: SJC		NORTH RIM – GRAND CANYON NATIONAL PARK
	TECH. REVIEW: BASH		
	DATE: AUG 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	13 OF 36

MATCHLINE, SEE SHEET 082C103

NORTH RIM CAMPGROUND LOOP

EXIST MH E-203
N: 13148412.16
E: 1327656.36

LINE M

MH E-204
N: 13148280.67
E: 1327746.96

CONSTRUCTION TURNAROUND LOOP

CONSTRUCTION TURNAROUND LOOP

EXIST MH H-201
N: 13148416.86
E: 1327901.15

LINE F

MH F-200
N: 13148956.41
E: 1328042.77

LIMITS OF CONSTRUCTION (TYP.)

MH F-201
N: 13148696.04
E: 1328004.15

MH G-200
N: 13148010.56
E: 1327765.22

MH E-205
N: 13148293.93
E: 1327900.99

LINE H

MH H-202
N: 13148344.39
E: 1327964.76

EXIST MH E-206
N: 13148226.01
E: 1328007.46

MH E-207
N: 13148203.92
E: 1328097.64

NORTH RIM CAMPGROUND LOOP

LINE G

LINE J

MH G-201
N: 13147987.18
E: 1328009.51

MH E-208
N: 13148133.87
E: 1328170.42

EXIST LIFT STATION 2
N: 13148144.79
E: 1328189.65

MATCHLINE, SEE SHEET 082C105

GENERAL NOTES:

A. CONSTRUCTION TRUCKS ONLY DRIVE INSIDE OF CONSTRUCTION LIMITS, AND FOLLOW CONSTRUCTION TRUCK PATH ON EXISTING ROAD. SEEDING AND MULCHING SHOULD BE IN DISTURBED AREA INSIDE OF CONSTRUCTION LIMITS.
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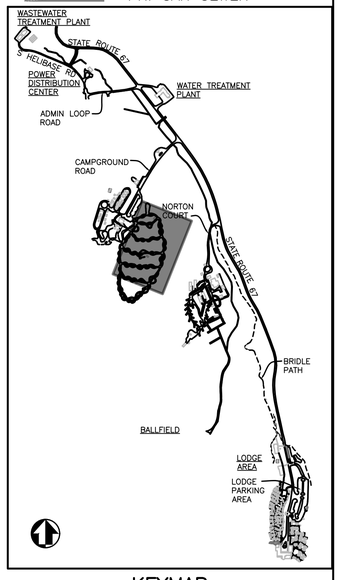


KEYMAP

40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 082 C104	TITLE OF SHEET SITEWORK GESC PLAN – REGION 2 – B NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 36

MH C-300
N: 13148257.21
E: 1328743.61

TP-EDR
N: 13147958.70
E: 1328904.87

LINE C

LINE A

WRANGLER RD

NORTH RIM DORMITORY

NORTON CT

EXIST MH C-301
N: 13148074.59
E: 1328578.84

MH A-301
N: 13147948.37
E: 1328654.62

EXIST MH B-300
N: 13147711.82
E: 1328886.77

EXIST MH B-301
N: 13147579.62
E: 1328948.76

MH B-302
N: 13147549.96
E: 1328967.81

EXIST MH A-302
N: 13147883.16
E: 1328603.00

EXIST MH B-306
N: 13147713.68
E: 1328816.68

EXIST MH B-305
N: 13147654.47
E: 1328777.36

NORTON ST

MH C-302
N: 13148033.23
E: 1328495.95

CONSTRUCTION
TURNAROUND LOOP

MH D-302
N: 13147712.31
E: 1328632.70

EXIST MH A-303
N: 13147799.81
E: 1328536.34

LINE AC

LINE B

NORTON CT

INDUSTRY RD

EXIST MH E-301
N: 1314742127
E: 1328925.07

LIMITS OF
CONSTRUCTION
(TYP.)

40' (TYP.)

LINE CE

EXIST MH C-303
N: 13147899.36
E: 1328428.26

EXIST MH B-303
N: 13147463.74
E: 1328782.80

EXIST MH E-300
N: 13147349.65
E: 1328859.61

LINE D

EXIST MH D-303
N: 13147568.32
E: 1328690.91

MH A-304
N: 13147742.82
E: 1328499.05

EXIST MH B-304
N: 13147421.65
E: 1328763.96

MH D-304
N: 13147321.71
E: 1328816.00

EXIST SEWAGE AND RECYCLED WATER DRAIN VALVES N: 13148032.85
E: 1328264.30

MATCHLINE, SEE SHEET 082C104

LINE M

CONCESSIONAIRE RESIDENCE RD

SPRUCE RD

CO-10
N: 13147330.62
E: 1328682.28

EXIST CO-8
N: 13147315.84
E: 1328855.20

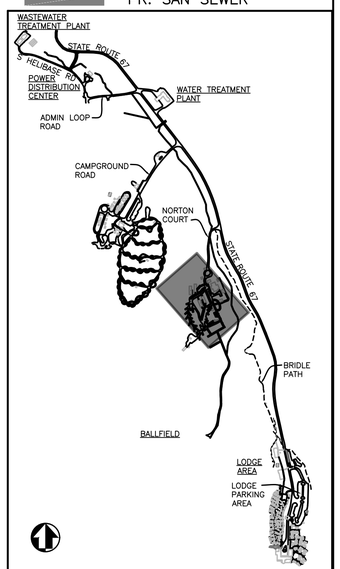
MH A-305
N: 13147533.07
E: 1328538.19

GENERAL NOTES:

A. CONSTRUCTION TRUCKS ONLY DRIVE INSIDE OF CONSTRUCTION LIMITS, AND FOLLOW CONSTRUCTION TRUCK PATH ON EXISTING ROAD. SEEDING AND MULCHING SHOULD BE IN DISTURBED AREA INSIDE OF CONSTRUCTION LIMITS.
B. SEE VOLUME 0 FOR EROSION CONTROL DETAILS.
C. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

LEGEND:

- CF — CONSTRUCTION FENCING 01/002C509
- SF — SILT FENCE 01/002C510
- SCL — SEDIMENT CONTROL LOG 03/002C510
- VTC — VEHICLE TRACKING CONTROL 01/002C511
- CWA — CONCRETE WASH AREA 02/002C511
- LOC — LIMITS OF CONSTRUCTION
- FLOW ARROW
- CONSTRUCTION TRUCK PATH
- PR. SAN SEWER



KEYMAP

40 0 40 80

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

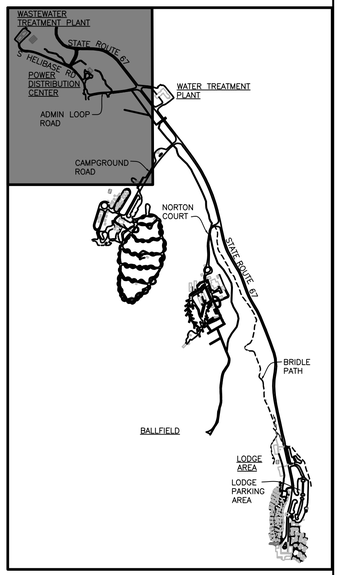
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 082 C105	TITLE OF SHEET SITEWORK GESC PLAN – REGION 3 NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 36

LEGEND:

- LINE A SEE 084C103
- LINES AB & AC SEE 084C107
- LINE C SEE 084C104 & C105
- LINES D & DE SEE 084C106
- LINE F SEE 084C107
- LINE DA SEE 084C102
- LINE DC SEE 084C102
- LINE M SEE 084C101

NOTES:

1. AREAS FOR ASPHALT REPLACEMENT OR REPAIR (APPROXIMATELY 17,625 SF FOR REGION 1) ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:
2. AREAS FOR GRAVEL REPLACEMENT OR REPAIR (APPROXIMATELY 1,895 SF FOR REGION 1) ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:
3. AREAS FOR SIDEWALK REPLACEMENT OR REPAIR (APPROXIMATELY 1,010 SF FOR REGION 1) ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:



KEYMAP

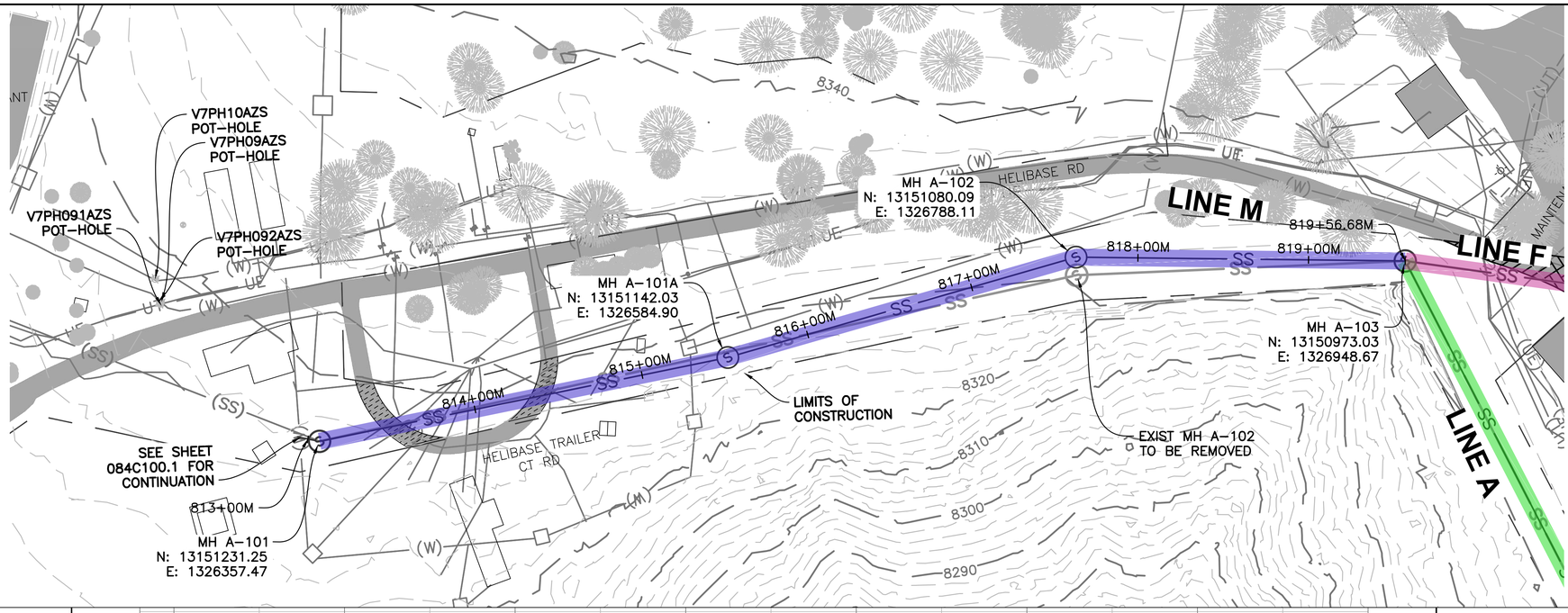
80 0 80 160 1" = 80'

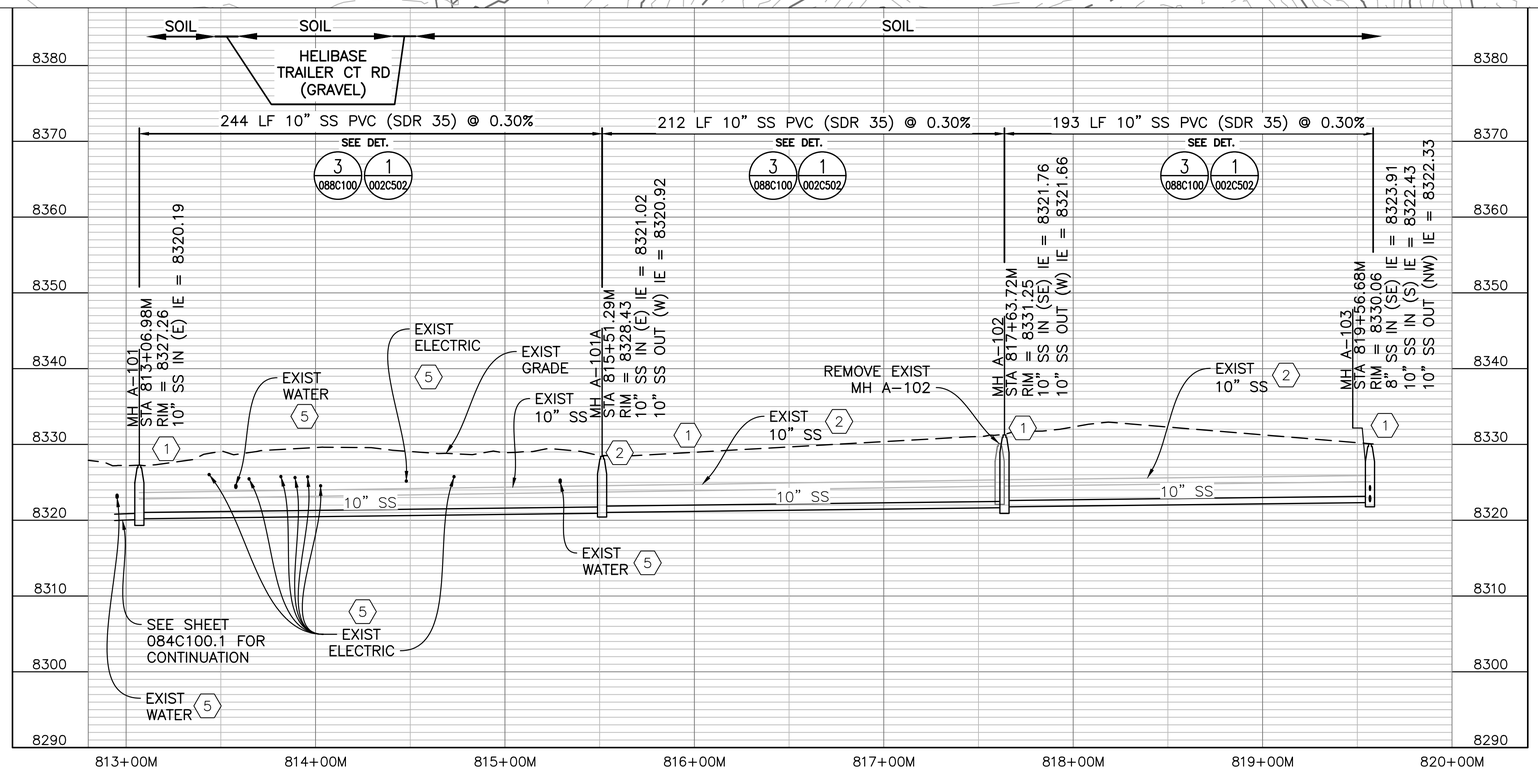
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C100	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN KEY MAP NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 36





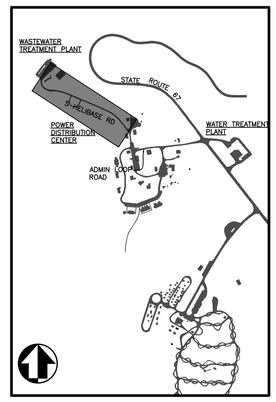
LINE M

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2–INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

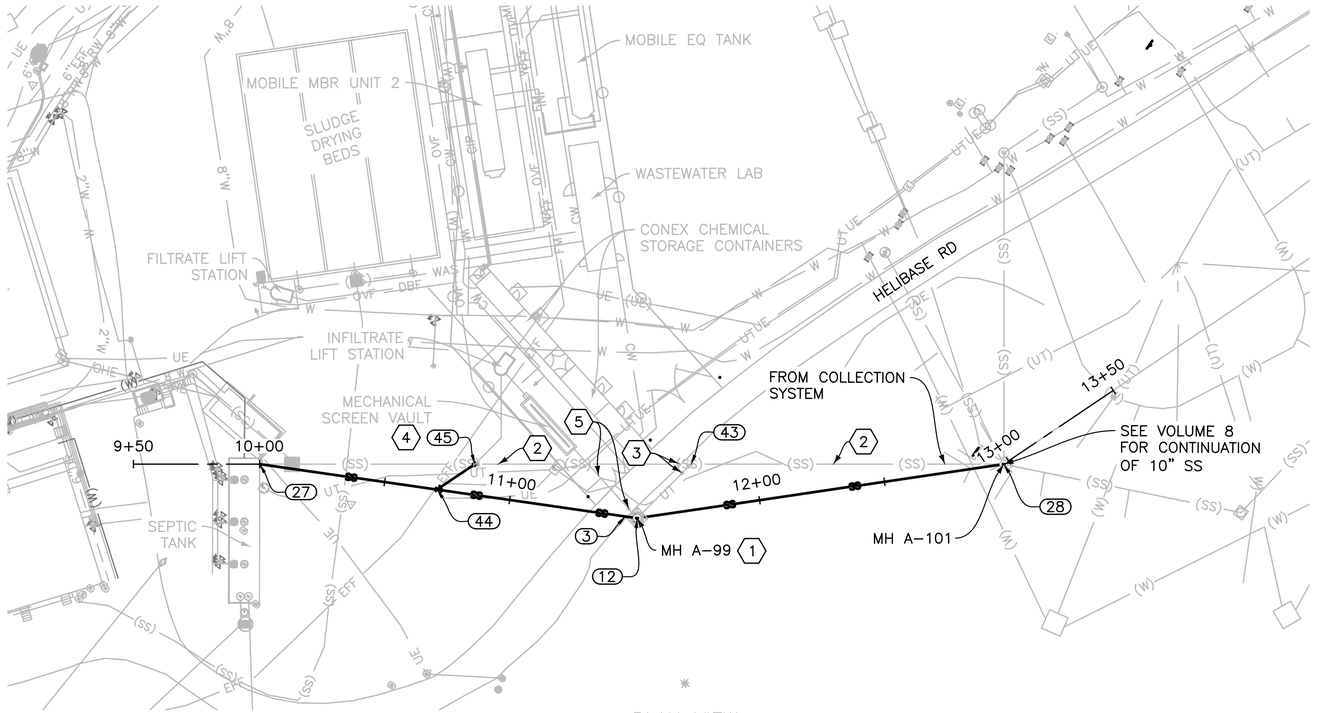


REGION 1 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 5/21/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C101	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINE M NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 17 of 36



PLAN VIEW

SCALE 1"=30'

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
3	13151341.03	1326253.32	PLUG VALVE
12	13151336.85	1326255.49	SSMH
27	13151470.77	1326185.80	CONNECT TO EXISTING SEPTIC TANK
28	13151231.33	1326357.48	SSMH
43	13151331.93	1326285.35	10"X10" WYE
44	13151407.86	1326218.54	10"X8" WYE
45	13151401.51	1326235.46	8" 11.25° HORIZONTAL BEND

GENERAL NOTES:

A. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

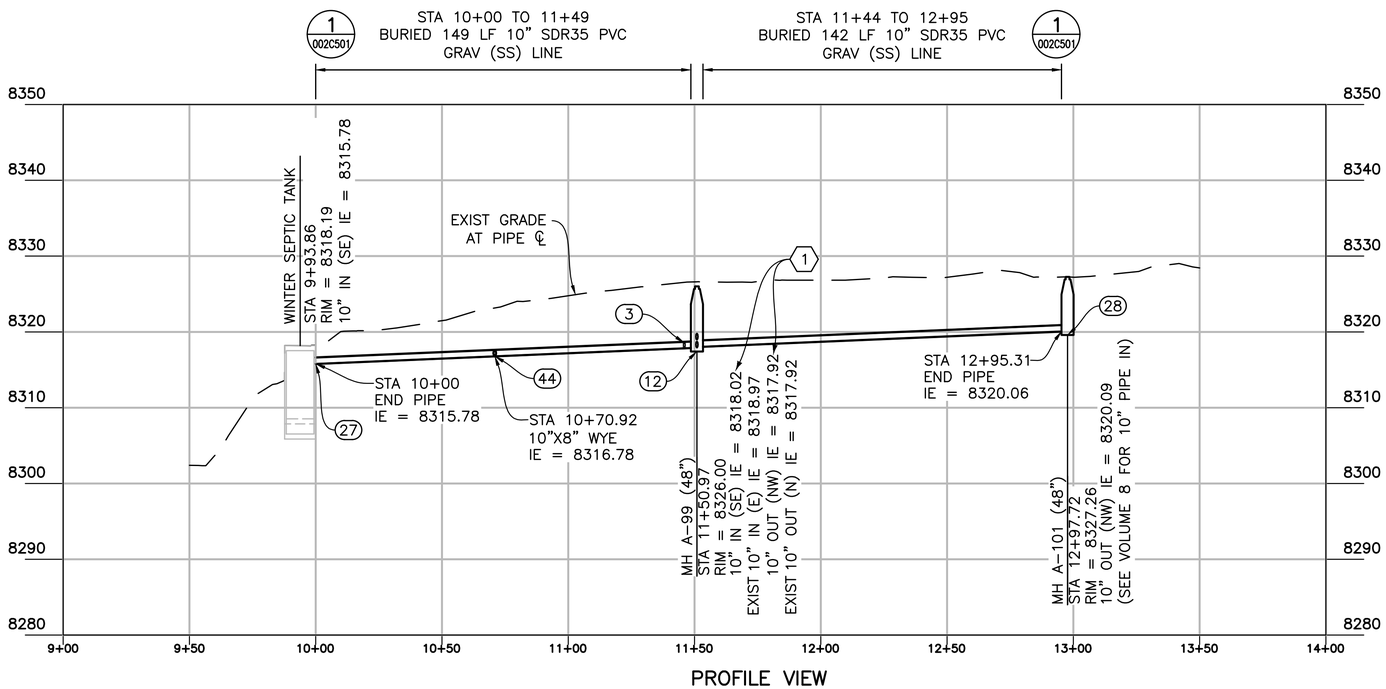
B. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

C. SURVEY CONTROL LOCATIONS AND POINT TABLE, POTHOLE LOCATIONS AND POINT TABLE, AND TYPICAL DETAILS ARE LOCATED IN VOLUME 0.

D. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT LINEAR FOOT MEASUREMENT.

KEY NOTES: (X)

1. CONTRACTOR SHALL CORE DRILL MH A-99 (EXISTING SEPTAGE RECEIVING MANHOLE) FOR 10" SDR 35 SANITARY SEWER.
2. EXISTING SANITARY SEWER SHALL BE PROTECTED IN PLACE DURING THE CONSTRUCTION OF NEW SEWER MAIN. UPON COMPLETION, EXISTING SANITARY SHALL BE ABANDONED IN PLACE.
3. UPON COMPLETION OF NEW SANITARY SEWER, ABANDON IN PLACE 10" PLUG VALVES. REMOVE VALVE STEMS AND BOX.
4. UPON COMPLETION OF NEW SANITARY SEWER, REMOVE 10"X8" WYE AND REPLACE WITH 8" 11.25° HORIZONTAL BEND AND CONNECT TO NEW SEWER MAIN AT NEW WYE.
5. EXIST 10" SS AND PLUG VALVE BETWEEN MH A-99 AND MECHANICAL SCREEN UNIT SHALL REMAIN AND NOT BE ABANDONED.



PROFILE VIEW

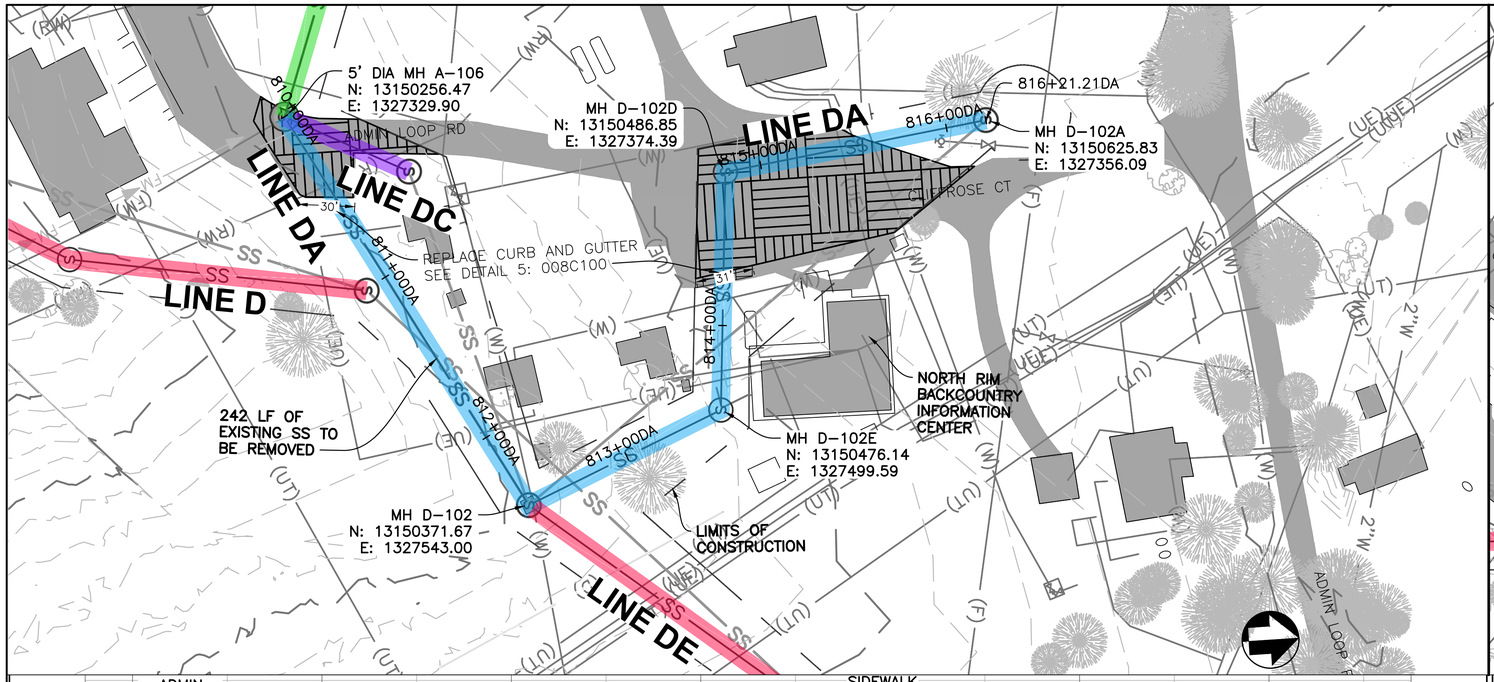
SCALE 1"=30'

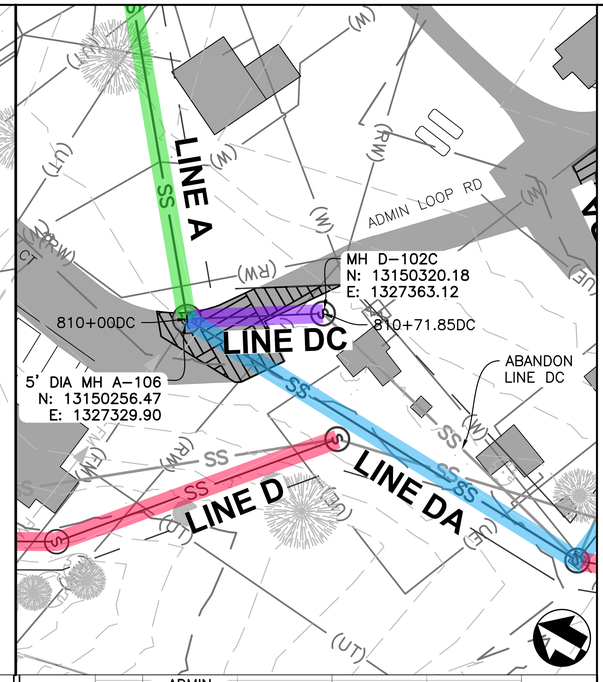
30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

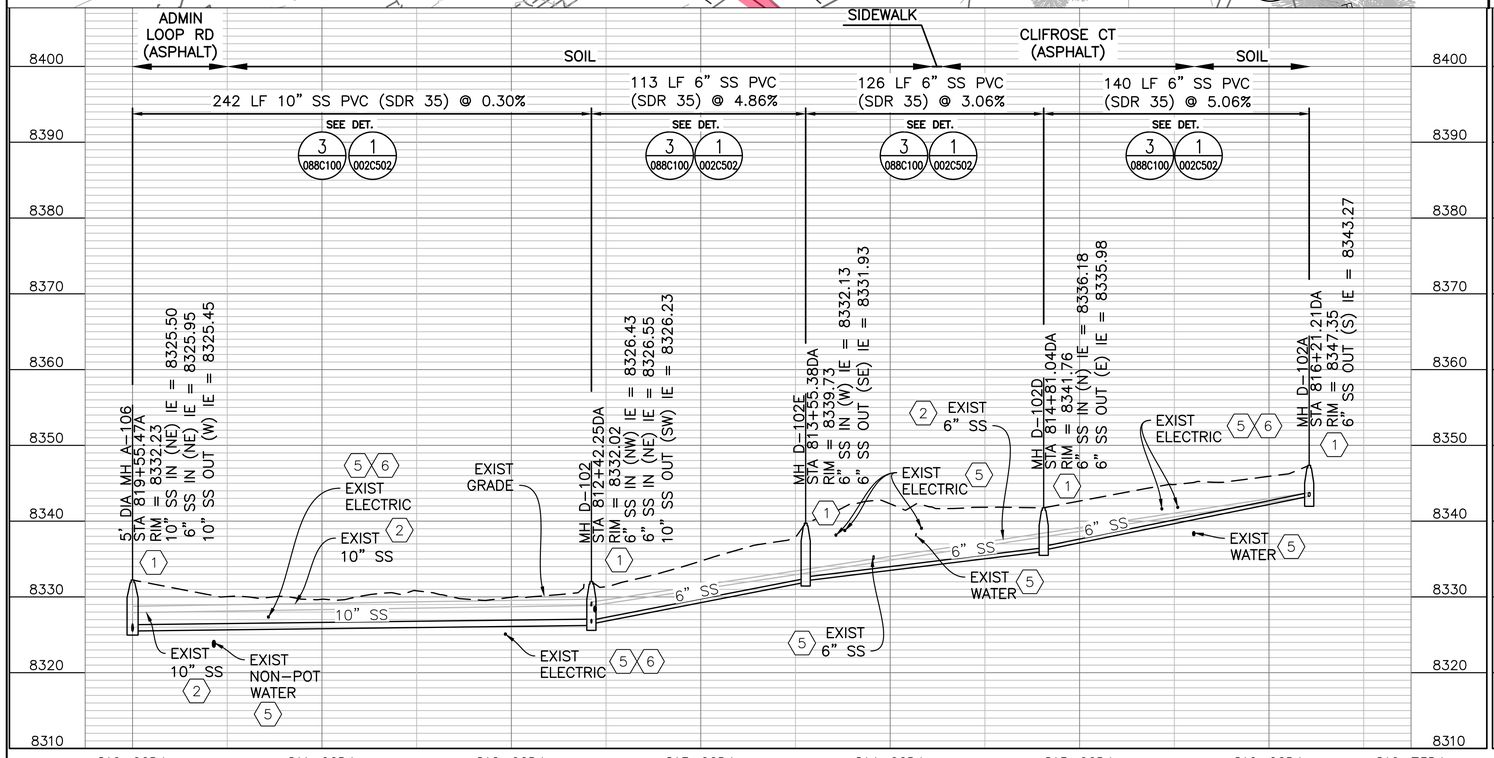
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/11/26 ARIZONA, U.S.A.

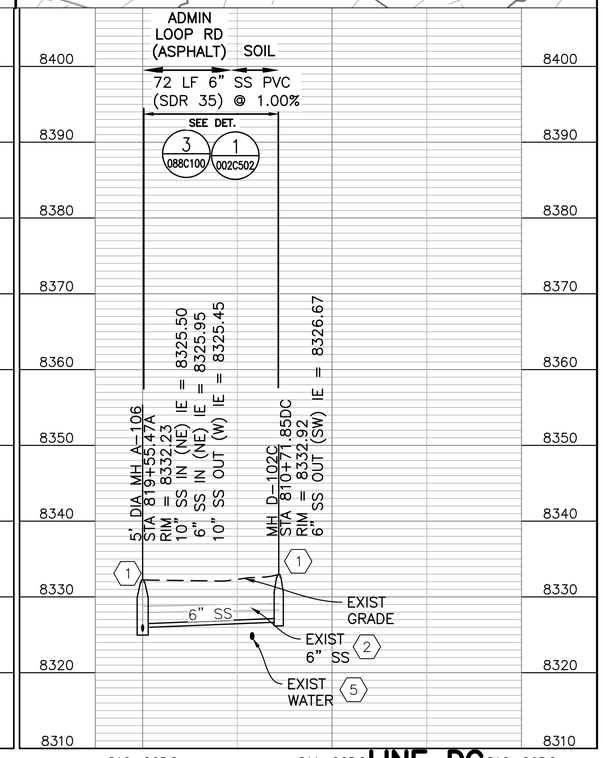
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: R. FREDERICKS DATE: MAY 2026	SUB SHEET NO. 084 C100.1	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE (10" SS) NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 35

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\084-C100.1.dwg







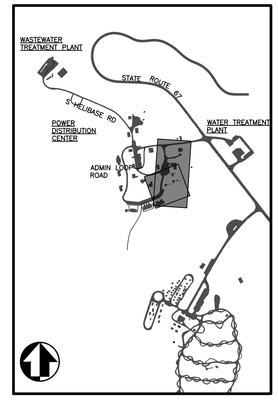


KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2-INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



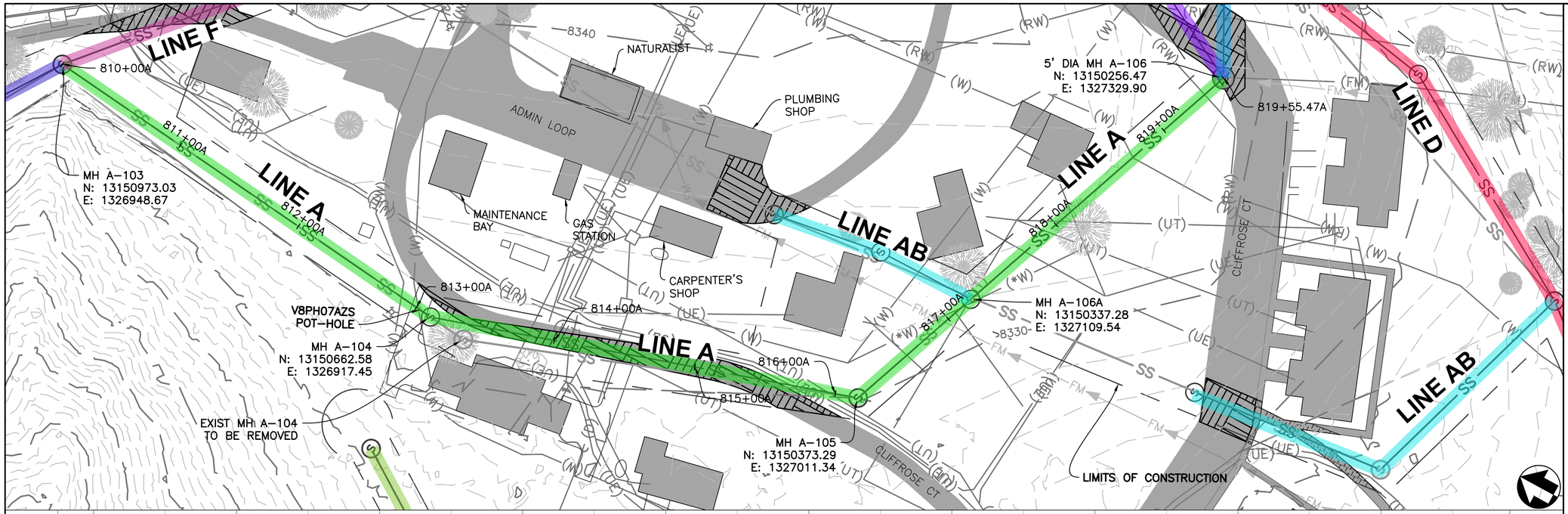
REGION 1 KEYMAP

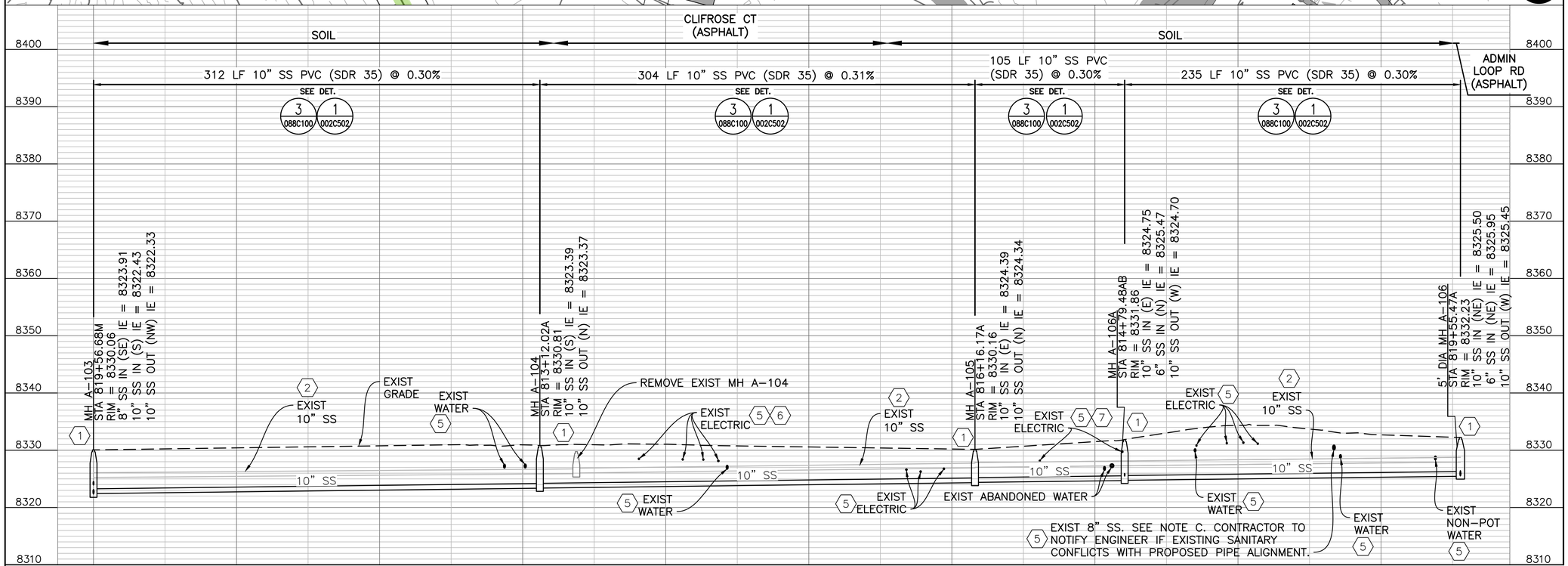
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

LINE DA

Registered Professional Engineer (Cert)
CERTIFICATE NO.
77465
BRIAN
STEPHENS-HOTOPP
Date Signed 8/21/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C102	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINES DA, DC NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 36	





LINE A

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
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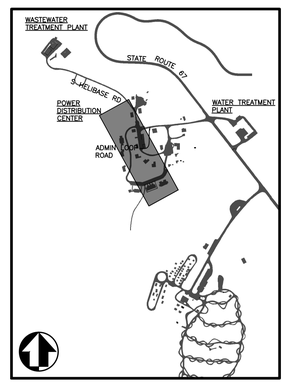


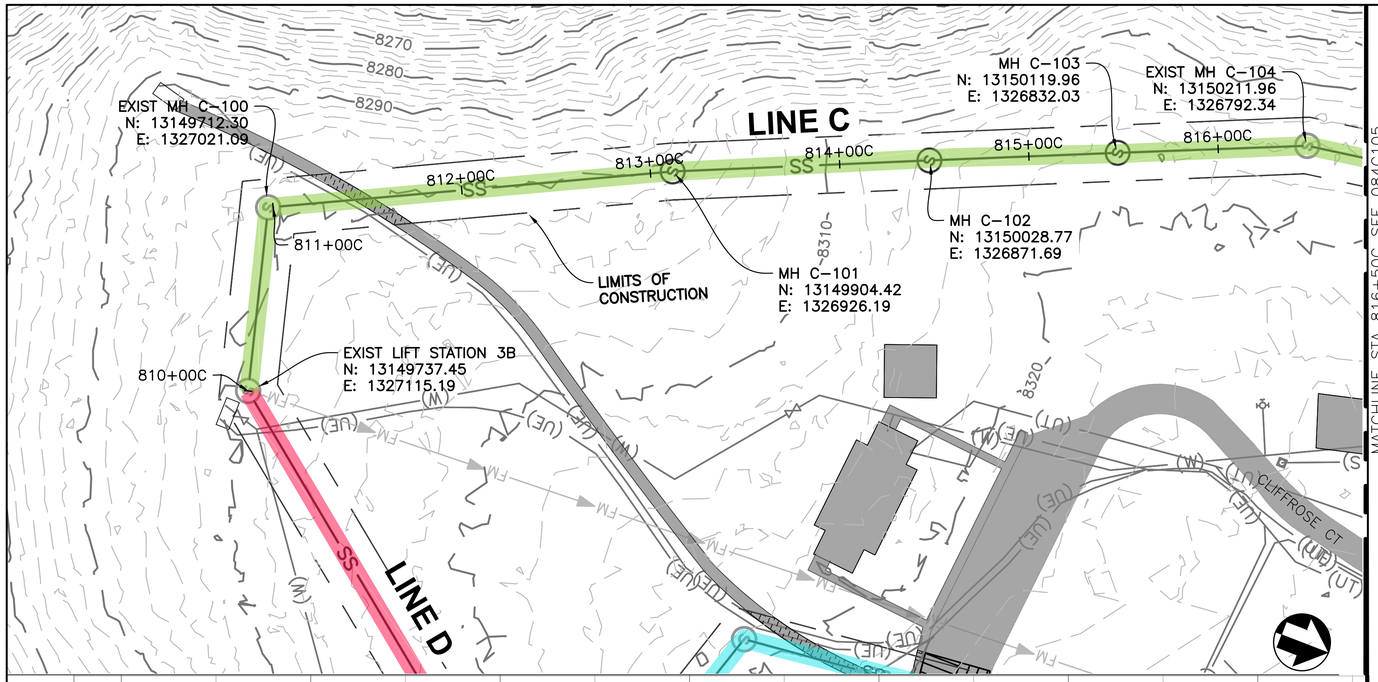
REGION 1 KEYMAP

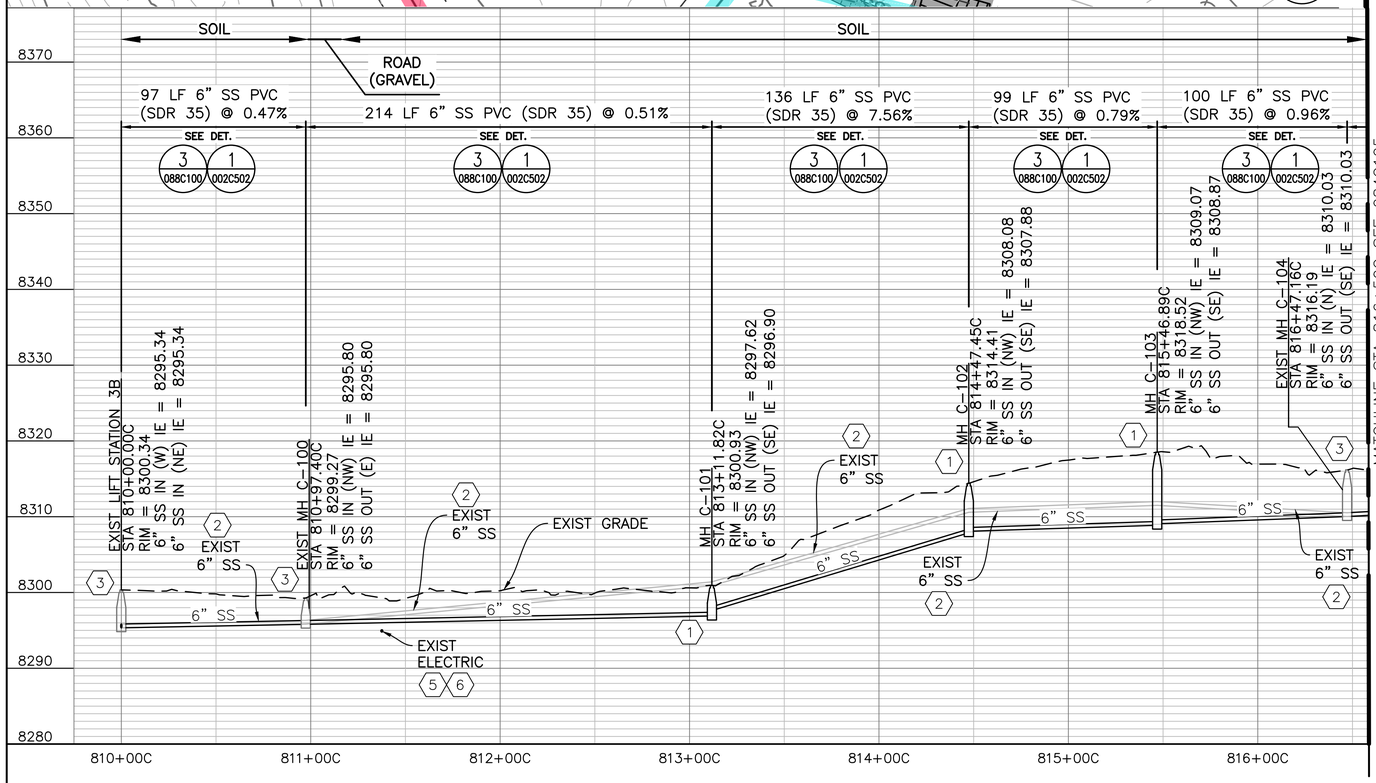
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C103	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINE A NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 20 OF 36





LINE C

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
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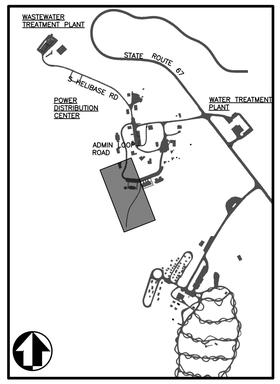
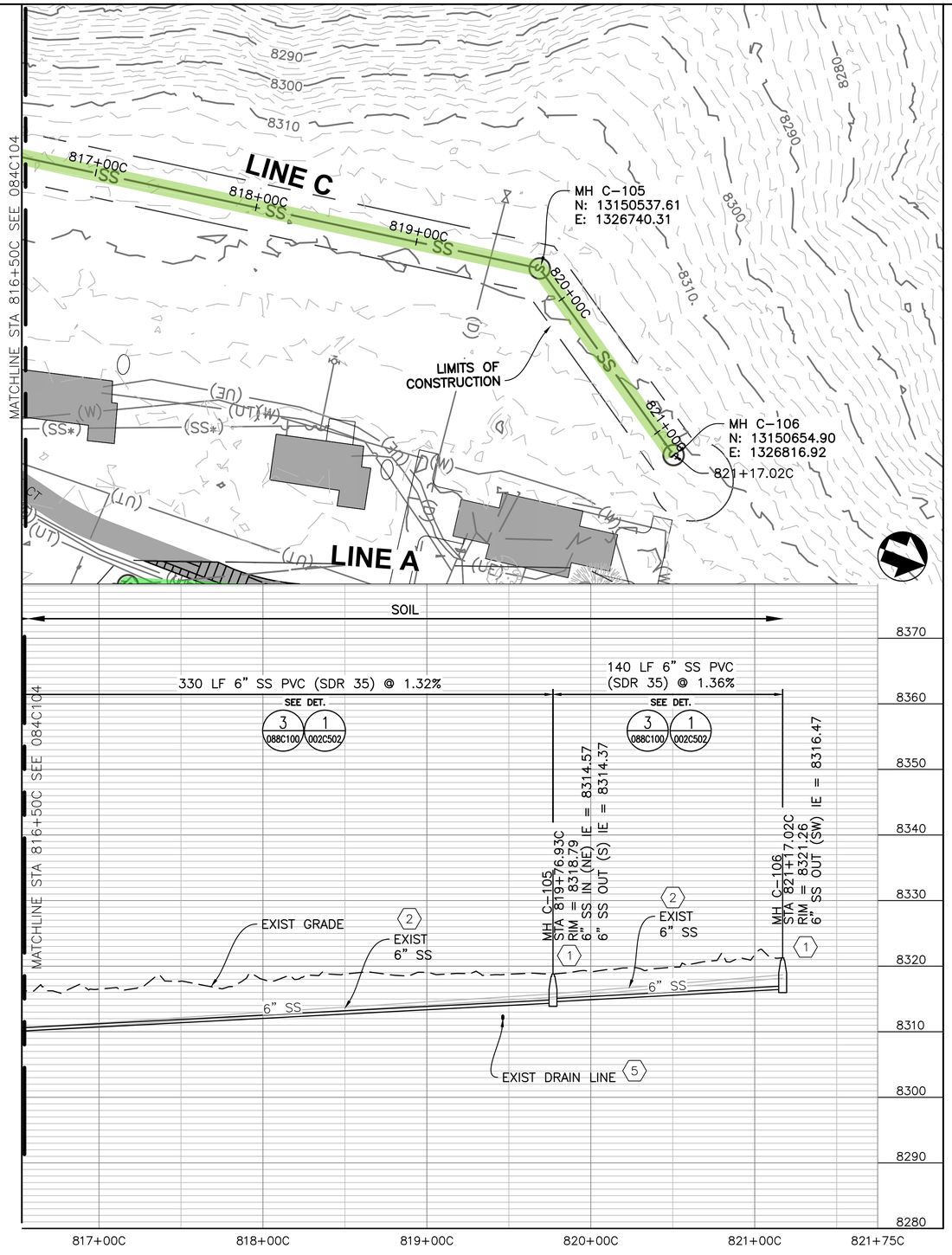


REGION 1 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Cert)
CERTIFICATE NO.
77465
BRIAN STEPHENS-HOTOPP
Date Signed 5/21/26
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C104	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINE C – 1 NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 36



LINE C

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
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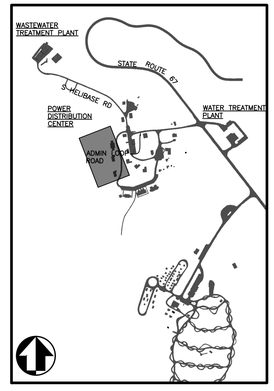
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2-INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

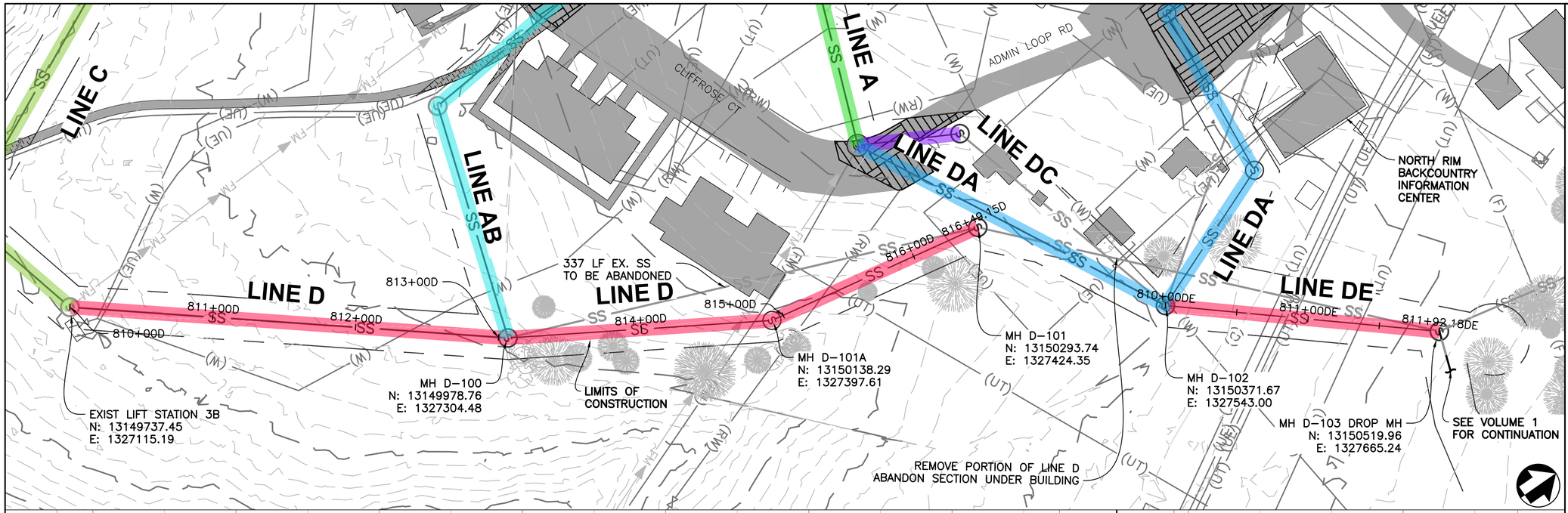


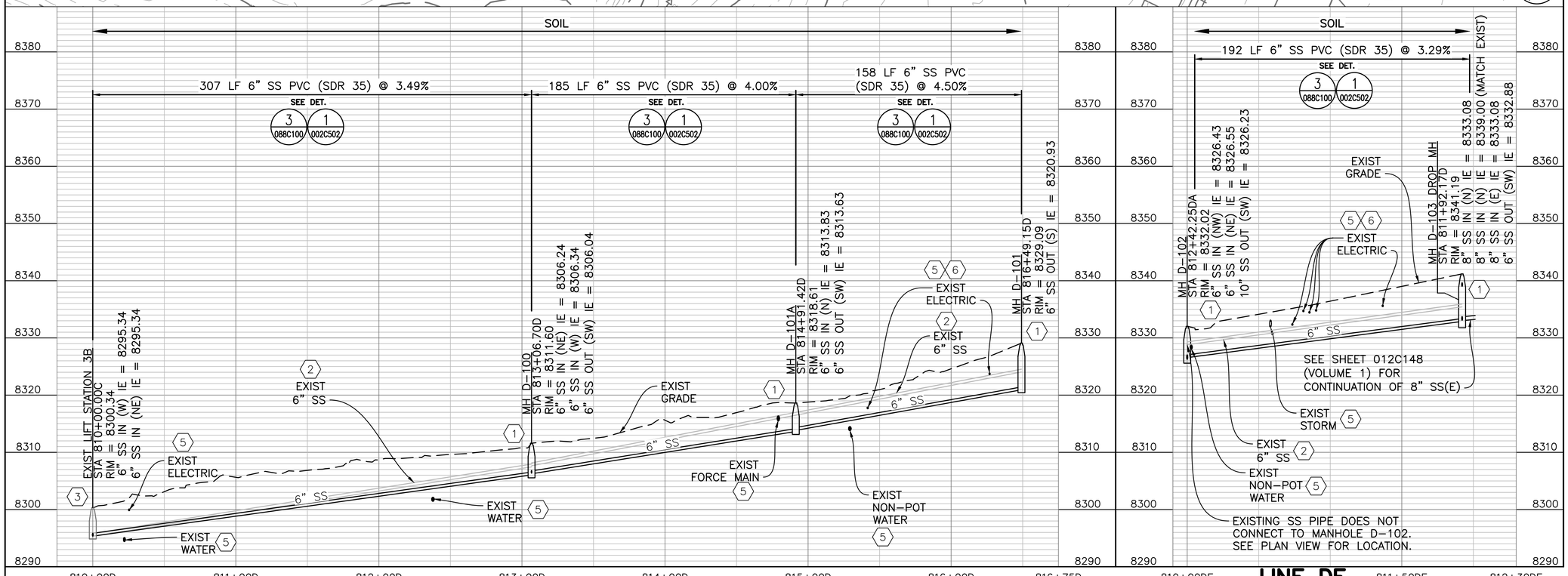
REGION 1 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C105	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINE C – 2 NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 36





LINE D

LINE DE

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
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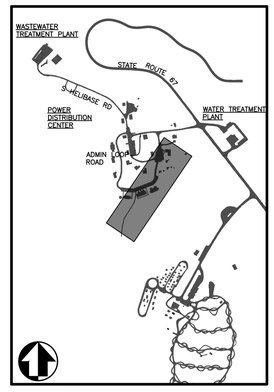
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



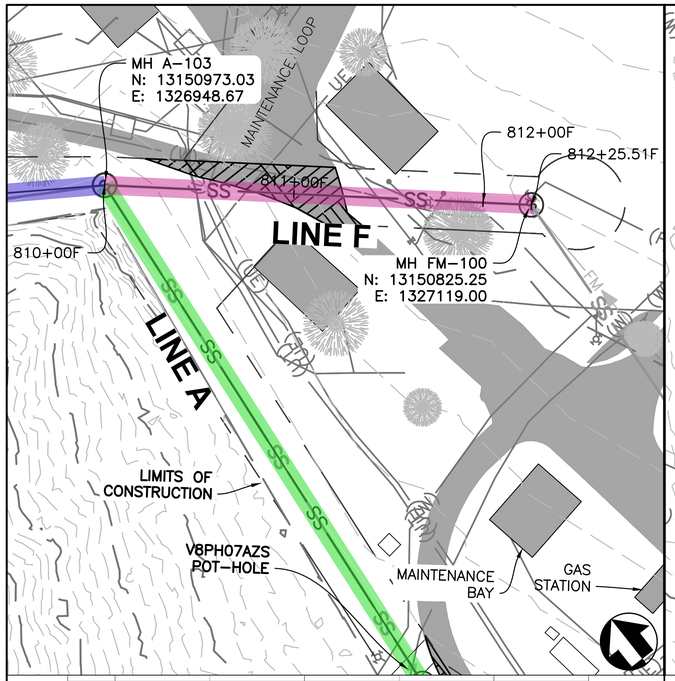
REGION 1 KEYMAP

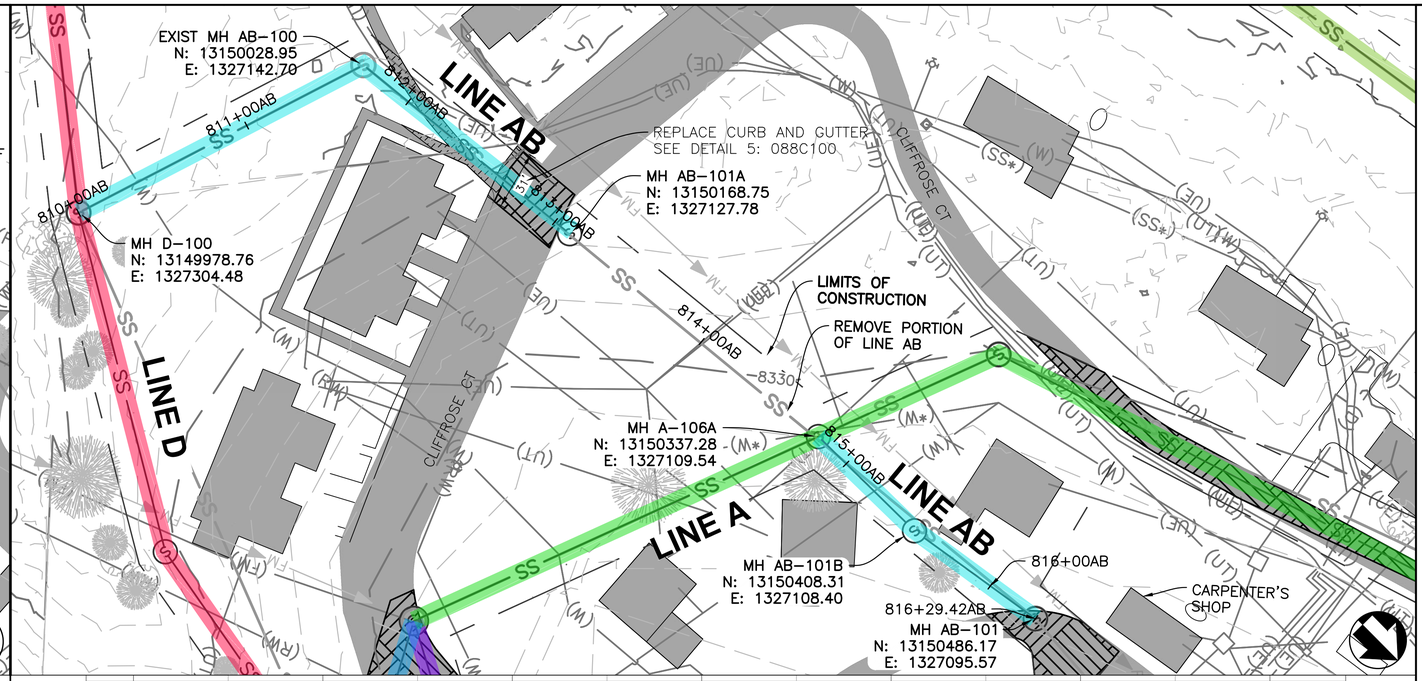
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

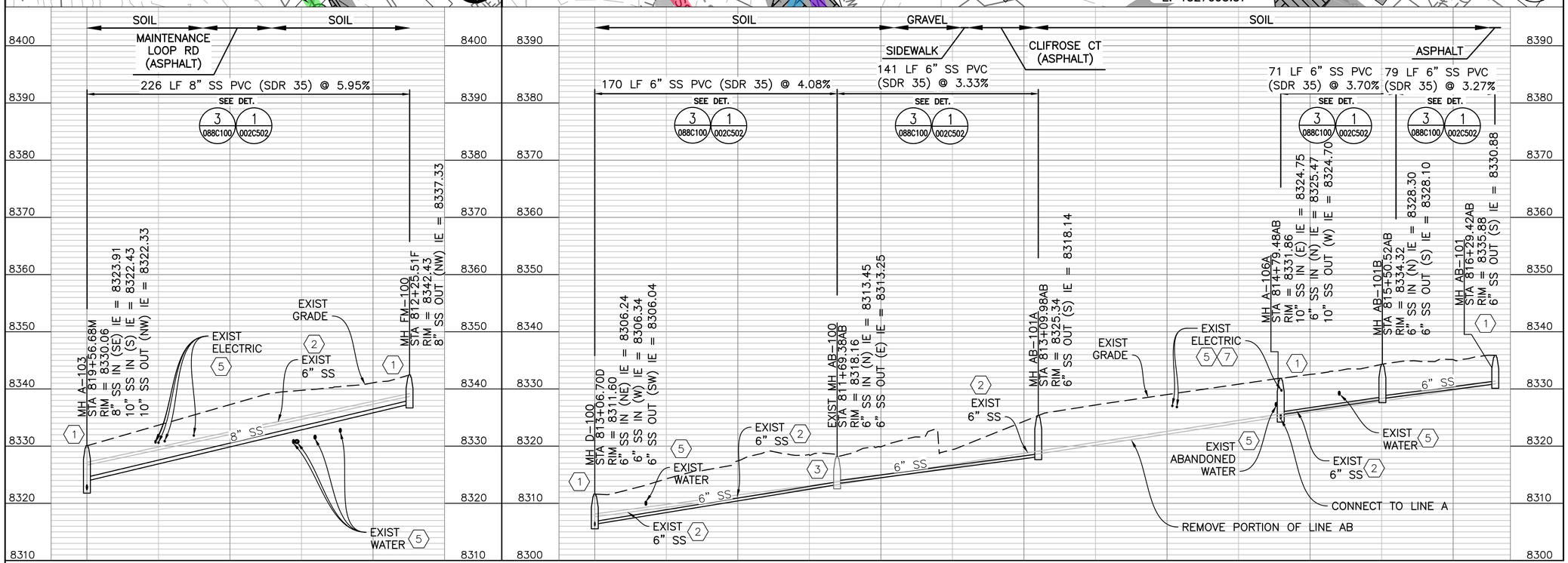
Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 9/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	084 C106	UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINES D, DE	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	CADD: SJC		NORTH RIM – GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 23 OF 36
	DATE: AUG 2026			







LINE F

LINE AB

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

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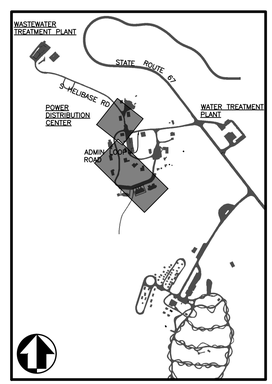


REGION 1 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 5/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 084 C107	TITLE OF SHEET UTILITIES – ADMIN AREA (REGION 1) SANITARY SEWER PLAN & PROFILE LINES F, AB NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 36

LINE E-A
LINE E-B
INN CABIN LOOP
LINE E
LINE B
LINE C
LINE A
LINE D
LINE M
LINE H
LINE F
LINE G
LINE J
SS
NORTH RIM GENERAL STORE
NORTH RIM CAMPGROUND LOOP
EXTERNAL FRAME CABIN LOOP
CAMPGROUND STORE ACCESS RD
GROUP CAMPSITE RD
CAMPGROUND RD
NORTH RIM CAMPGROUND LOOP
LIFT STATION 2

LEGEND:

LINE A SEE 085C102	
LINE B SEE 085C103	
LINE C SEE 085C104	
LINE D SEE 085C101	
LINE E SEE 085C103	
LINE E SERVICES	
LINE EA SEE 085C106	
LINE EB SEE 085C106	
LINE F SEE 085C104	
LINE G SEE 085C105	
LINE H SEE 085C103	
LINE J SEE 085C105	
LINE M SEE 085C101	

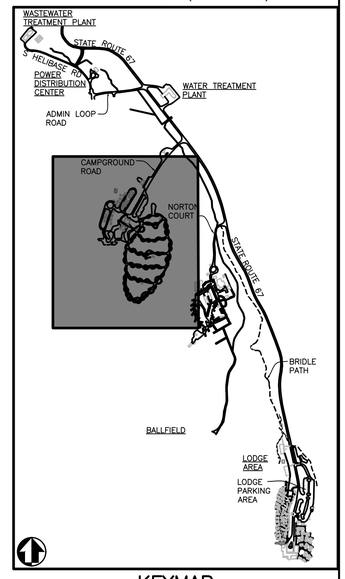
NOTES:

1. AREAS FOR ASPHALT REPLACEMENT OR REPAIR ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:

   APPROXIMATELY 15,755 SF OF ASPHALT REPAIR OR REPLACEMENT IS IDENTIFIED FOR CAMPGROUND AREA (REGION 2).

2. AREAS FOR GRAVEL REPLACEMENT OR REPAIR ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:

   APPROXIMATELY 8,595 SF OF GRAVEL REPAIR OR REPLACEMENT IS IDENTIFIED FOR CAMPGROUND AREA (REGION 2).



KEYMAP

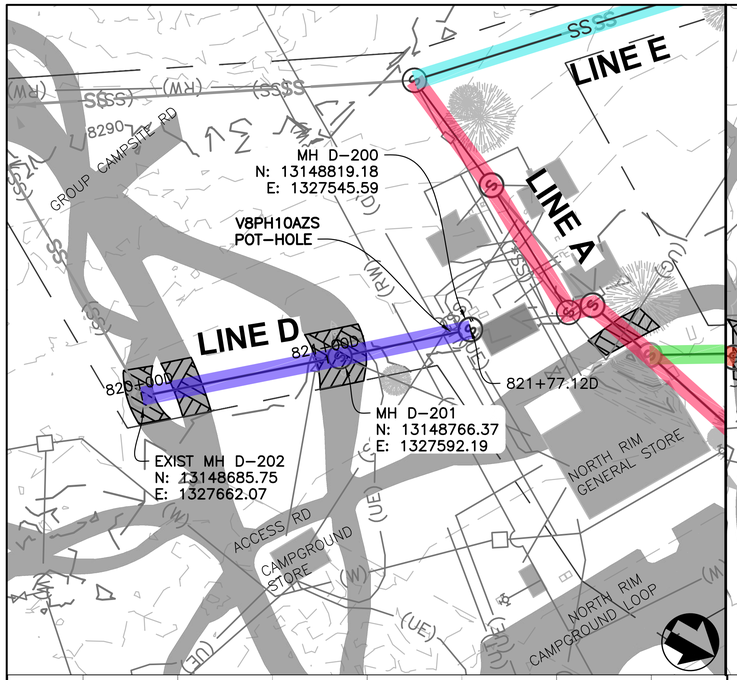
80 0 80 160

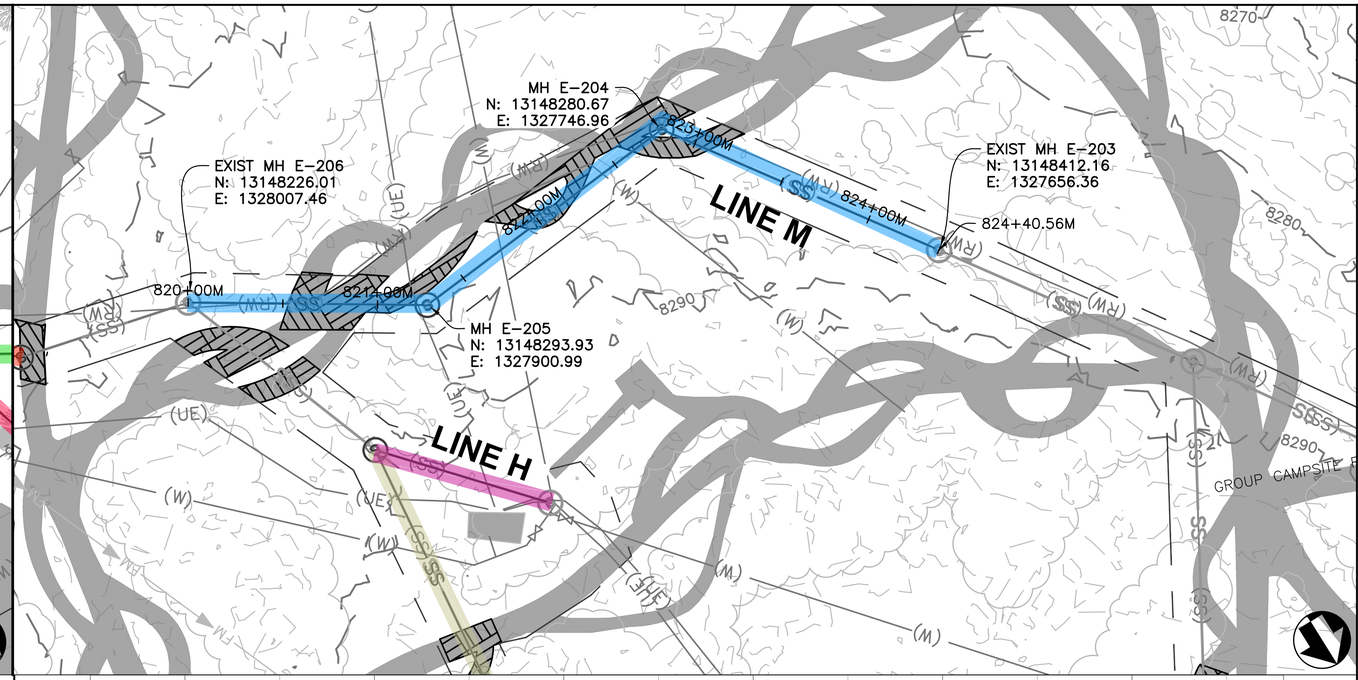
SCALE OF FEET 1" = 80'

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS-HOTOPP Not Signed 6/27/25 ARIZONA U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 085 C100	TITLE OF SHEET **UTILITIES – CAMPGROUND AREA (REGION 2)** **SANITARY SEWER PLAN KEY MAP** NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 36





KEY NOTES

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2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
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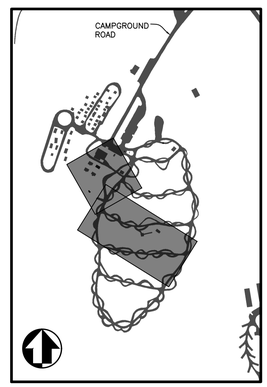
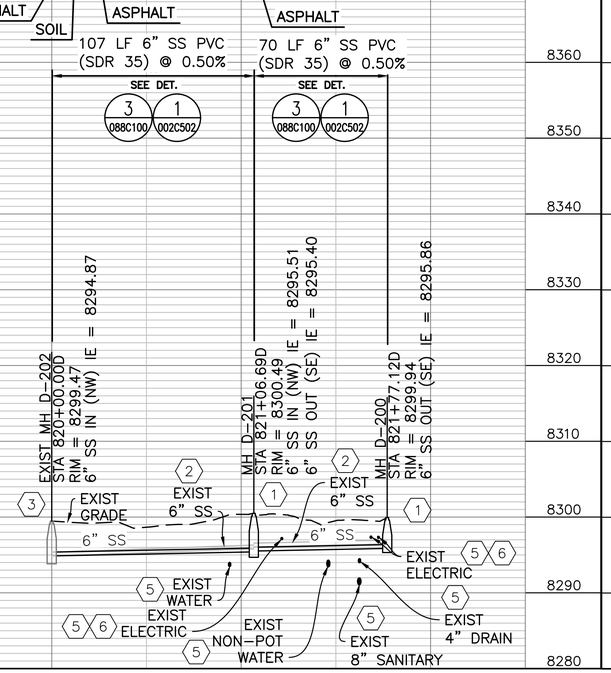
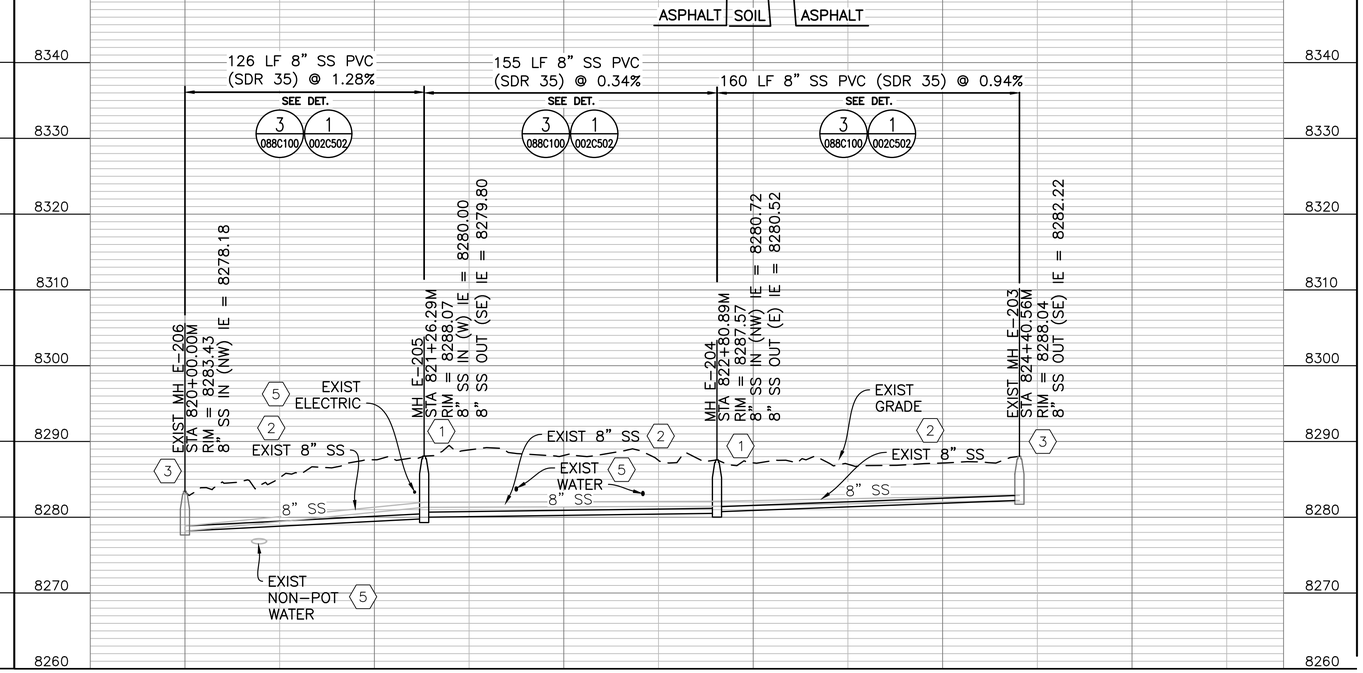


REGION 2 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET



LINE D



LINE M

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS–HOTOPP Date Signed 6/17/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

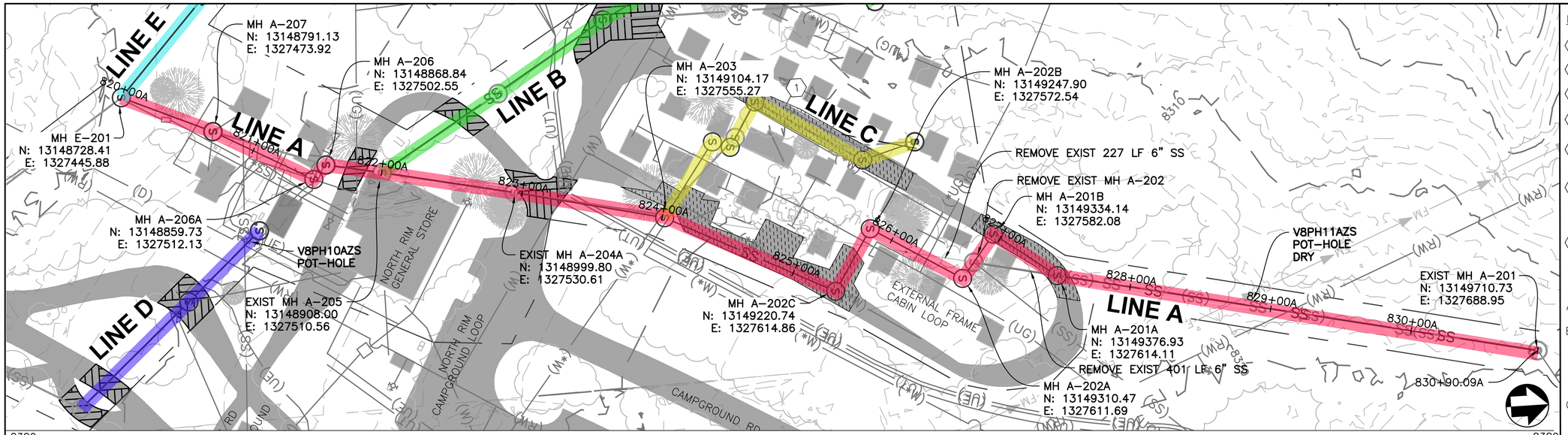
A/E FIRM
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO

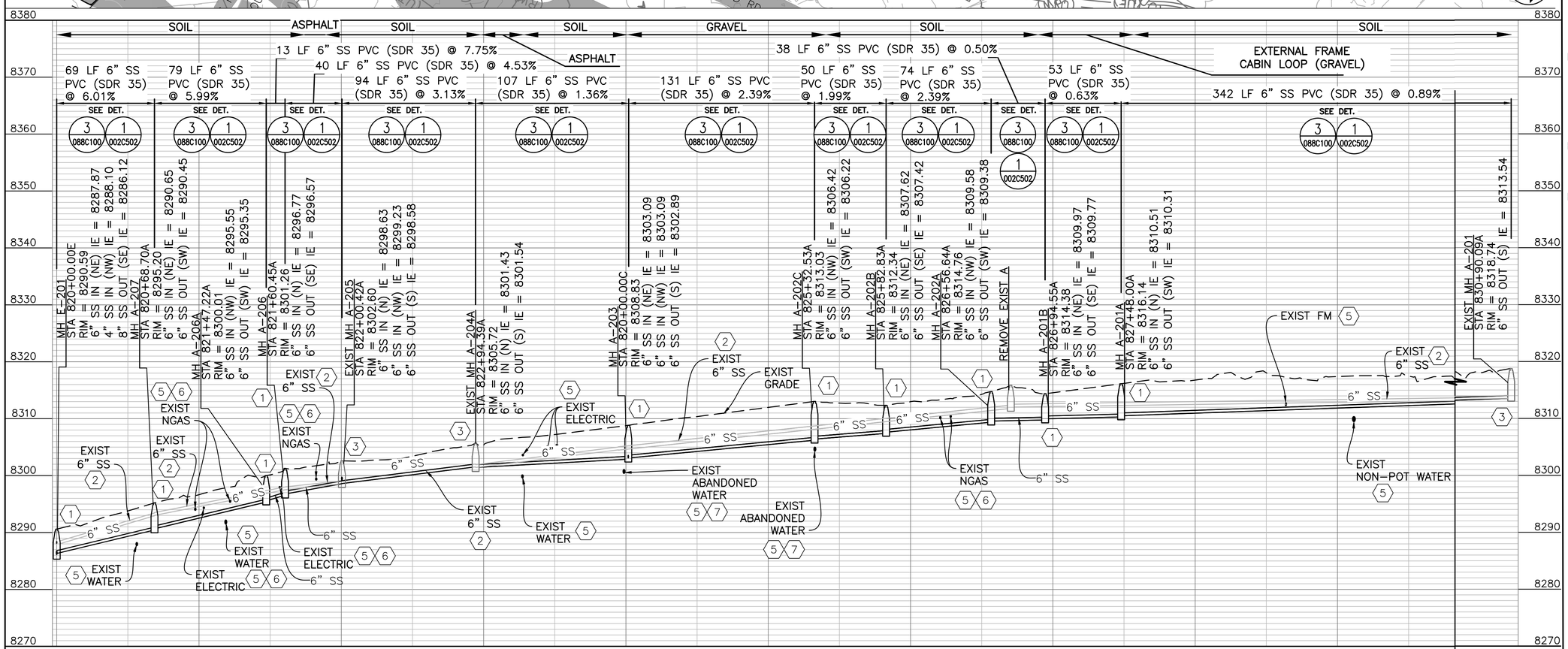
DESIGNED: KNF/SJC
DRAWN: SJC
TECH. REVIEW: BASH
DATE: AUG 2026

SUB SHEET NO. 085 C101

TITLE OF SHEET
UTILITIES – CAMPGROUND AREA (REGION 2)
SANITARY SEWER PLAN & PROFILE
LINES D, M
NORTH RIM – GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 26 OF 36





LINE A

KEY NOTES

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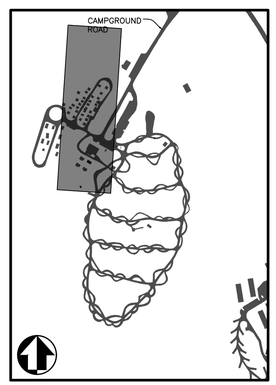


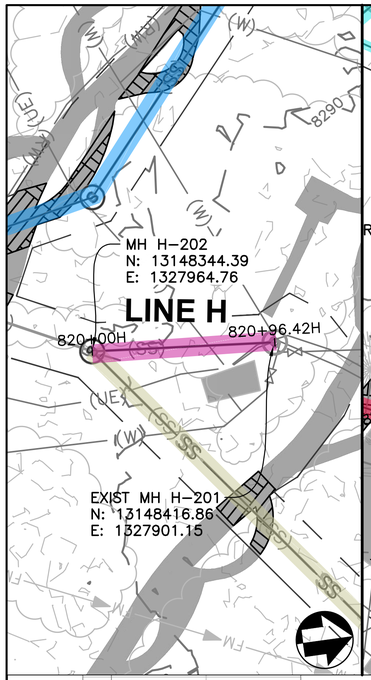
REGION 2 KEYMAP

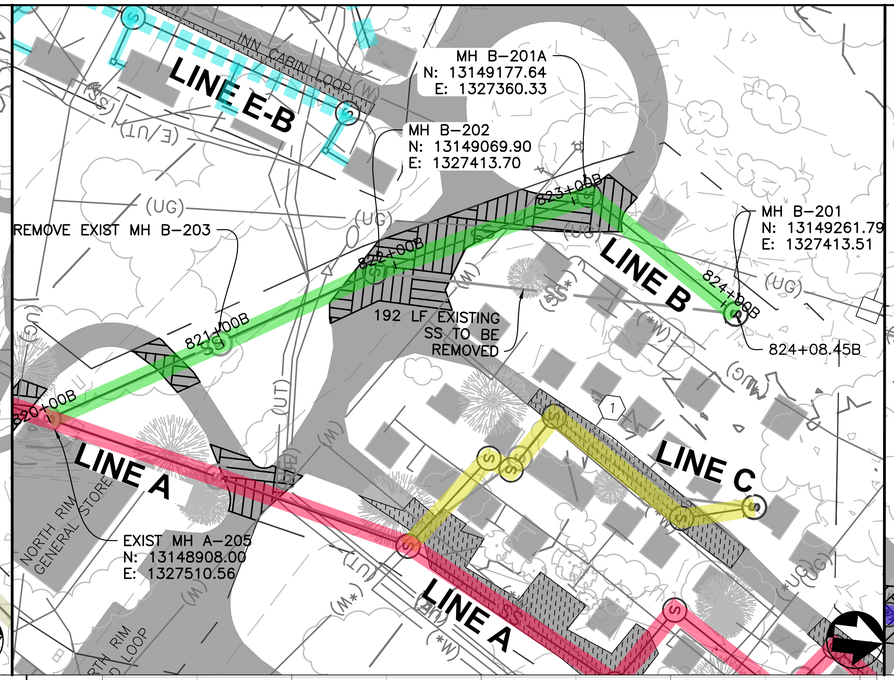
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

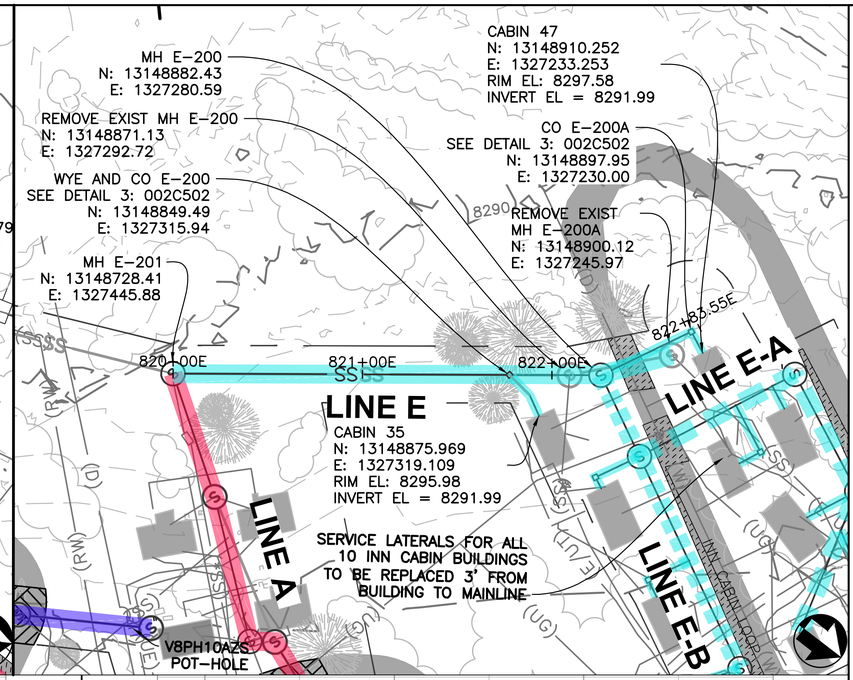
Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP 26th Signed 8/21/25 ARIZONA, U.S.A.

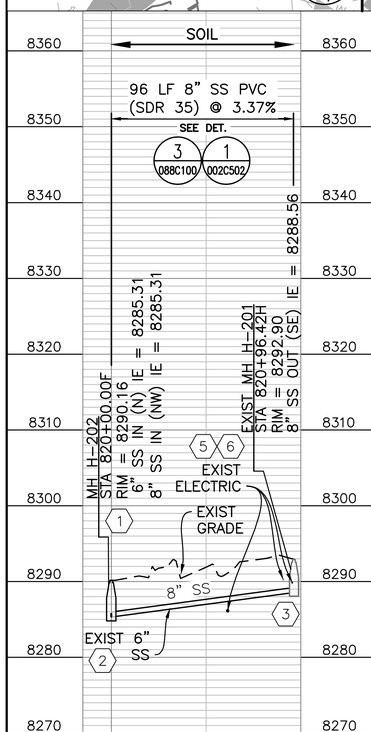
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	085 C102	UTILITIES – CAMPGROUND AREA (REGION 2) SANITARY SEWER PLAN & PROFILE LINE A
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DRAWN: SJC		NORTH RIM – GRAND CANYON NATIONAL PARK
	TECH. REVIEW: BASH		
	DATE: AUG 2026		

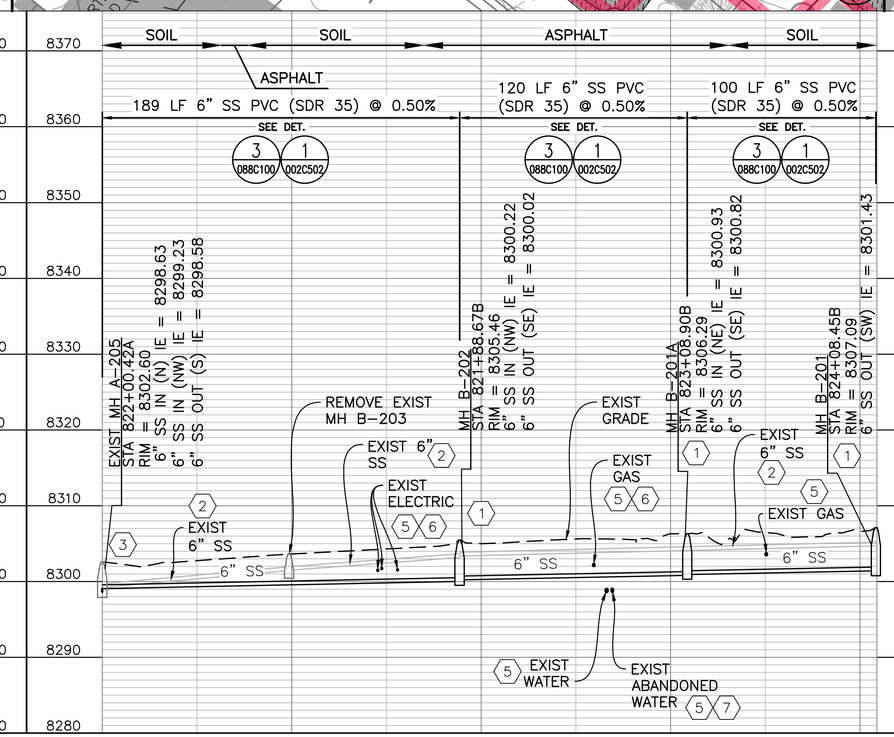
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	27 OF 36

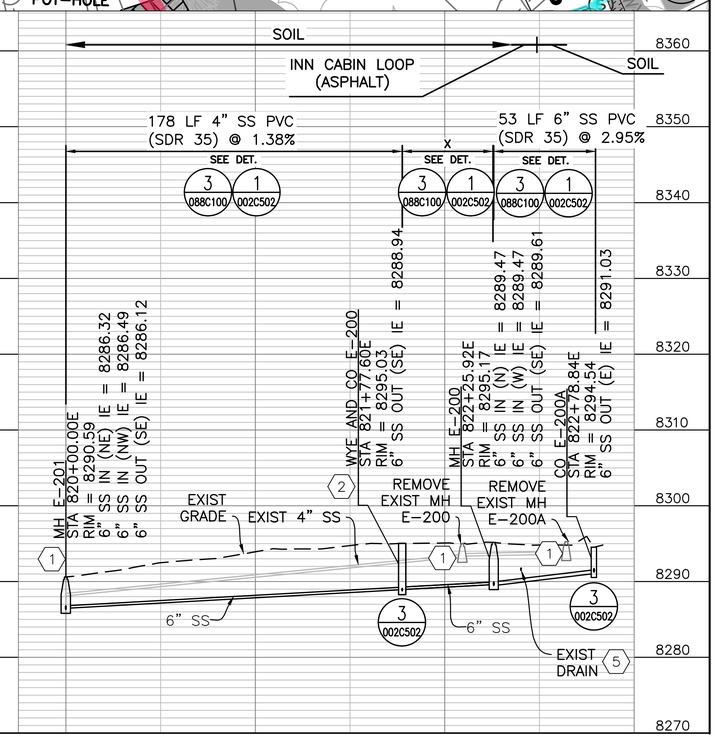












820+00H 821+00H **820+00B** 821+00B 822+00B 823+00B **824+00B** 820+00E 821+00E 822+00E 823+00E

LINE H **LINE B** **LINE E**

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
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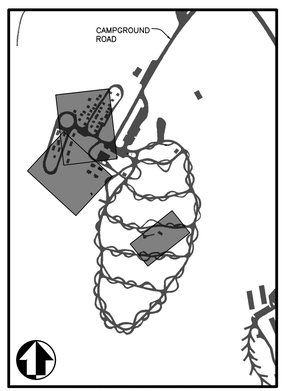


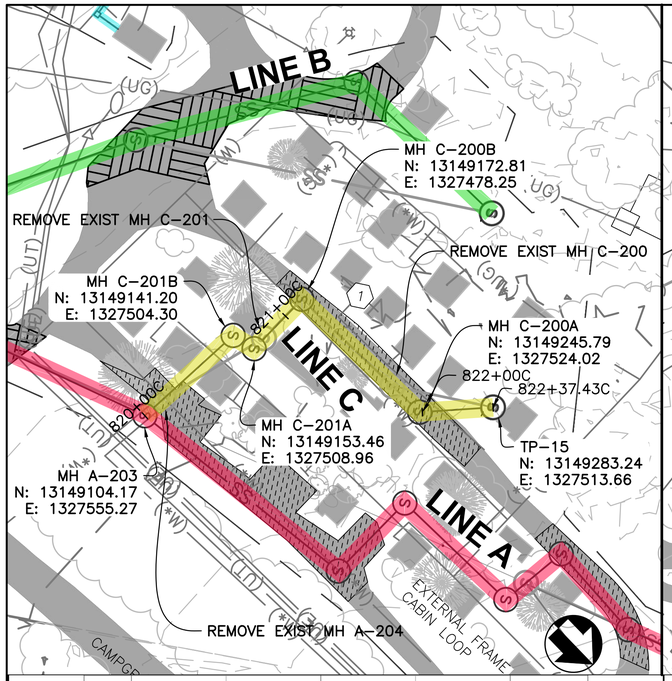
REGION 2 KEYMAP

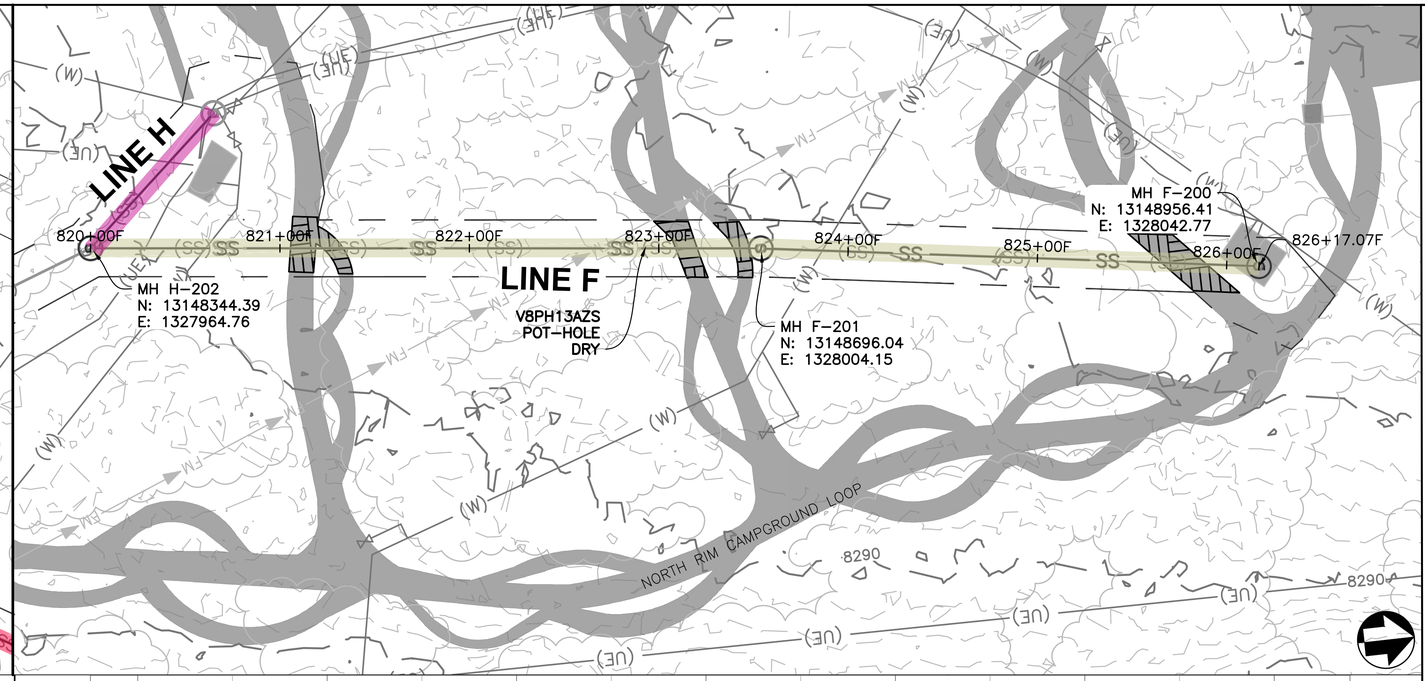
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

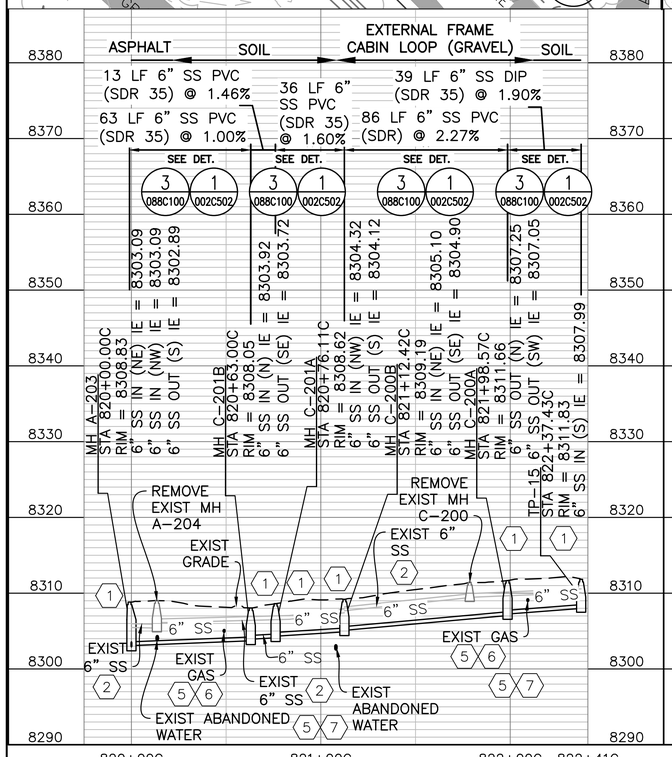
Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS–HOTOPP Date Signed 9/21/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

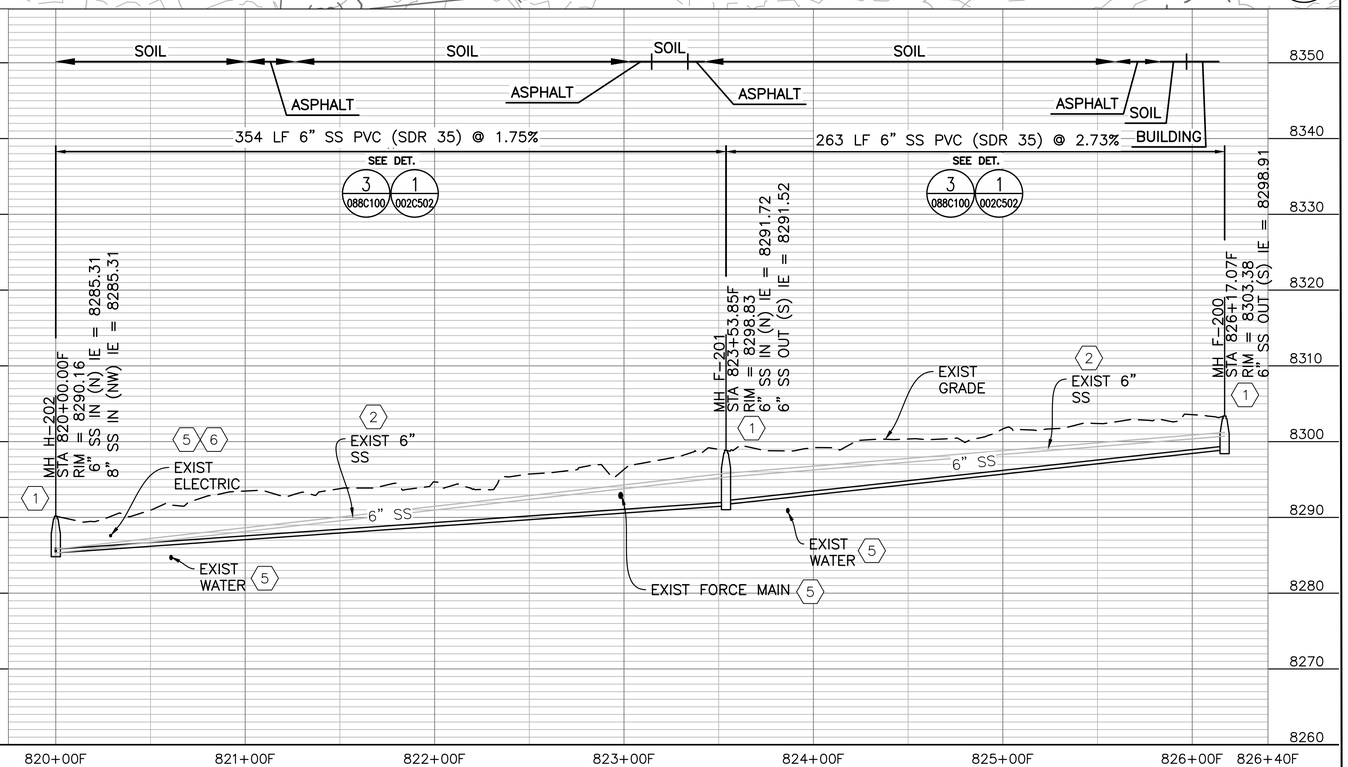
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 085 C103	TITLE OF SHEET UTILITIES – CAMPGROUND AREA (REGION 2) SANITARY SEWER PLAN & PROFILE LINES E, B, H NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 36







LINE C



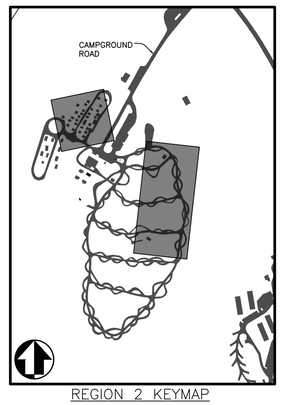
LINE F

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2-INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



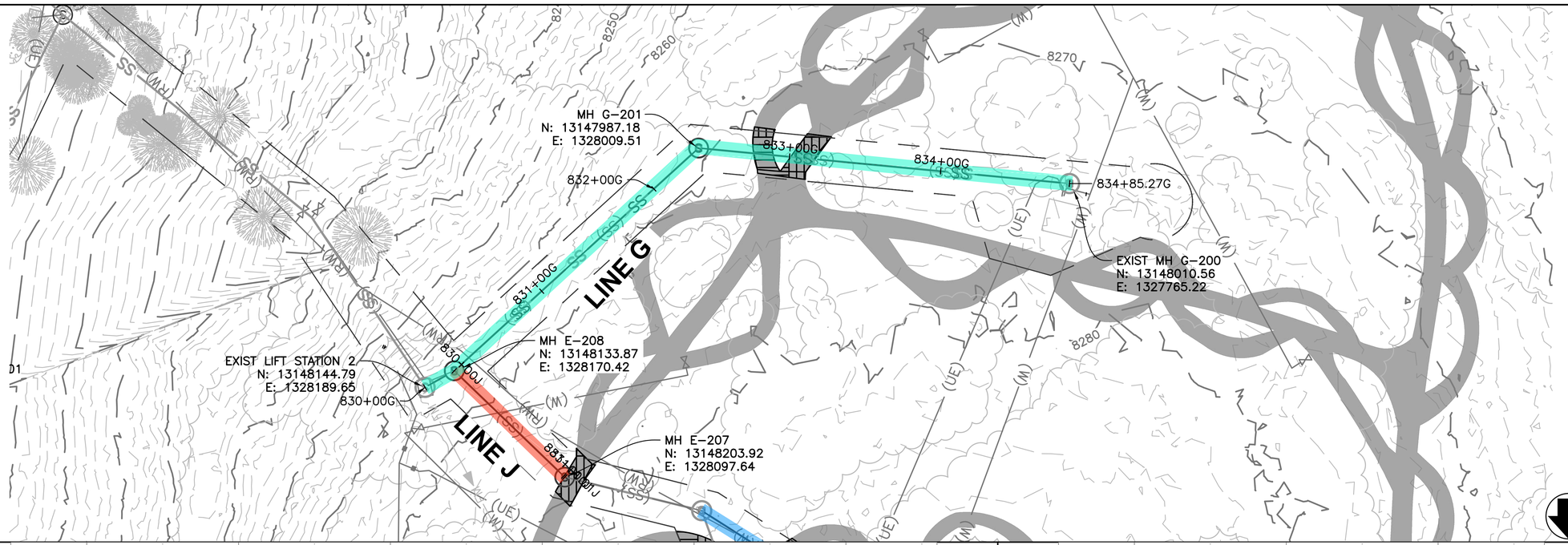
REGION 2 KEYMAP

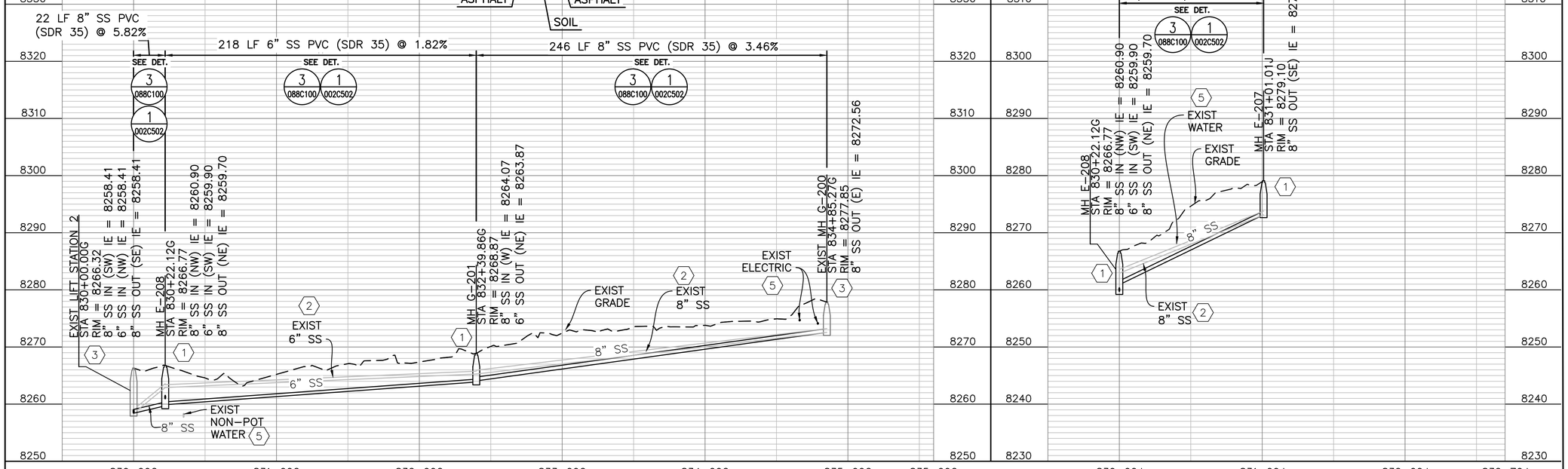
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 6/21/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED: KNF/SJC	SUB SHEET NO. 085 C104	TITLE OF SHEET: UTILITIES – CAMPGROUND AREA (REGION 2) SANITARY SEWER PLAN & PROFILE LINES C, F NORTH RIM – GRAND CANYON NATIONAL PARK
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	CADD: SJC		DRAWING NO. 113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	TECH. REVIEW: BASH		PMIS/PKG NO. 318719
	DATE: AUG 2026		SHEET 29 OF 36





LINE G

LINE J

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
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6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
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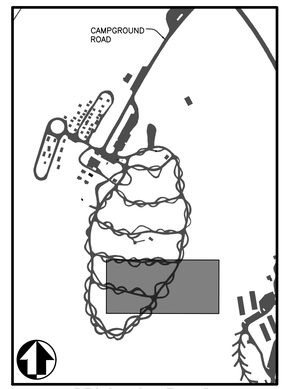
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



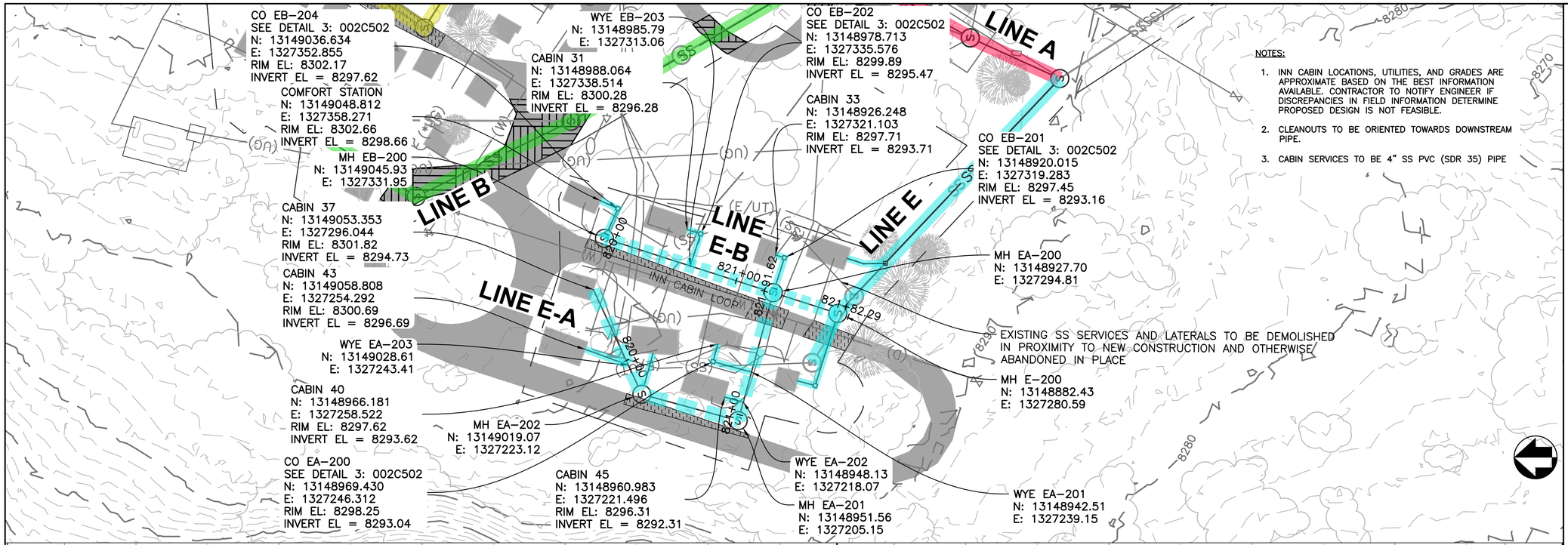
REGION 2 KEYMAP

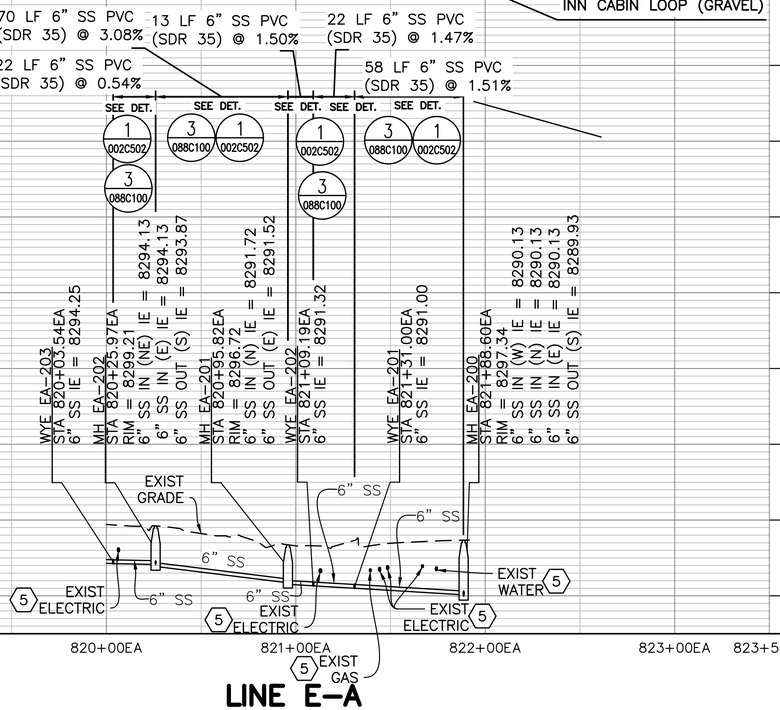
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

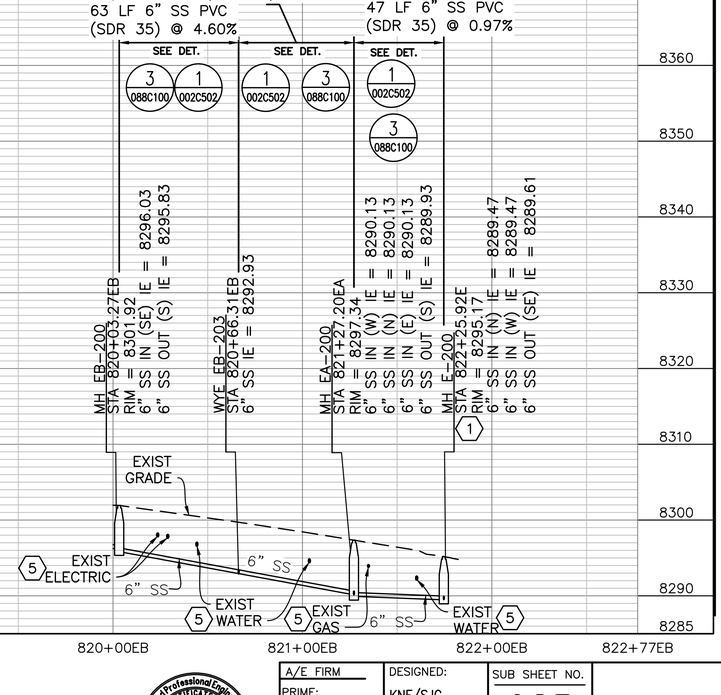
Registered Professional Engineer (Civil)
CERTIFICATE NO.
77465
BRIAN
STEPHENS-HOTOPP
Date Signed 8/21/26
ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 085 C105	TITLE OF SHEET UTILITIES – CAMPGROUND AREA (REGION 2) SANITARY SEWER PLAN & PROFILE LINES G, J NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 36





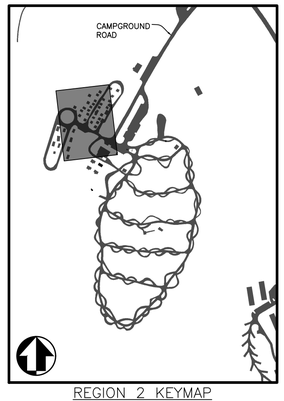


KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2–INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



REGION 2 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS–HOTOPP Date Signed 6/1/26 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED: KNF/SJC	SUB SHEET NO. 085 C106	TITLE OF SHEET: UTILITES – CAMPGROUND AREA (REGION 2) SANITARY SEWER PLAN & PROFILE LINES E–A , E–B NORTH RIM – GRAND CANYON NATIONAL PARK
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	CADD: SJC		DRAWING NO. 113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	TECH. REVIEW: BASH		PMIS/PKG NO. 318719
	DATE: AUG 2026		SHEET 31 OF 36

LEGEND:

LINE A SEE 086C102

LINE AC SEE 086C102

LINE B SEE 086C103

LINE C SEE 086C101

LINE D SEE 086C102

LINE M SEE 086C103

NOTES:

1. AREAS FOR ASPHALT REPLACEMENT OR REPAIR ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:

   APPROXIMATELY 25,100 SF OF ASPHALT REPAIR OR REPLACEMENT IS IDENTIFIED FOR CONCESSIONAIRE AREA (REGION 3).

2. AREAS FOR SIDEWALK REPLACEMENT OR REPAIR ARE IDENTIFIED ON THE PLANS WITH THE FOLLOWING SYMBOL:

   APPROXIMATELY 1,010 SF OF SIDEWALK REPAIR OR REPLACEMENT IS IDENTIFIED FOR CONCESSIONAIRE AREA (REGION 3).

CAMPGROUND AREA (REGION 2)

CONCESSIONAIRE AREA (REGION 3)

LINE M

LINE AC

LINE C

LINE A

LINE D

LINE B

LIFT STATION 3A

RESIDENCE RD

CONCESSIONAIRE

SPRUCE RD

DUSTY RD

NORTON CT

WRANGLER RD

EMPLOYEE DINING ROOM

NORTH RIM DORMITORY

WASTEWATER TREATMENT PLANT

STATE ROUTE 67

S HELIBASE RD

POWER DISTRIBUTION CENTER

WATER TREATMENT PLANT

ADMIN LOOP ROAD

CAMPGROUND ROAD

NORTON COURT

BALLFIELD

LODGE AREA

LODGE PARKING AREA

KEYMAP

80 0 80 160

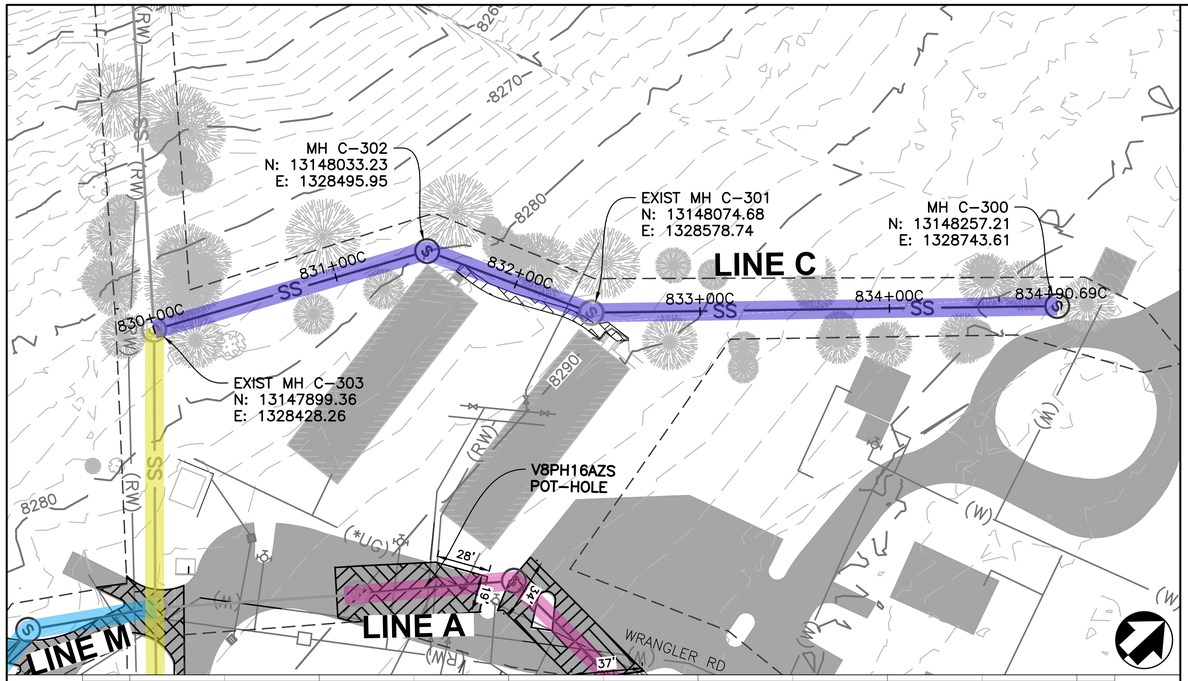
SCALE OF FEET

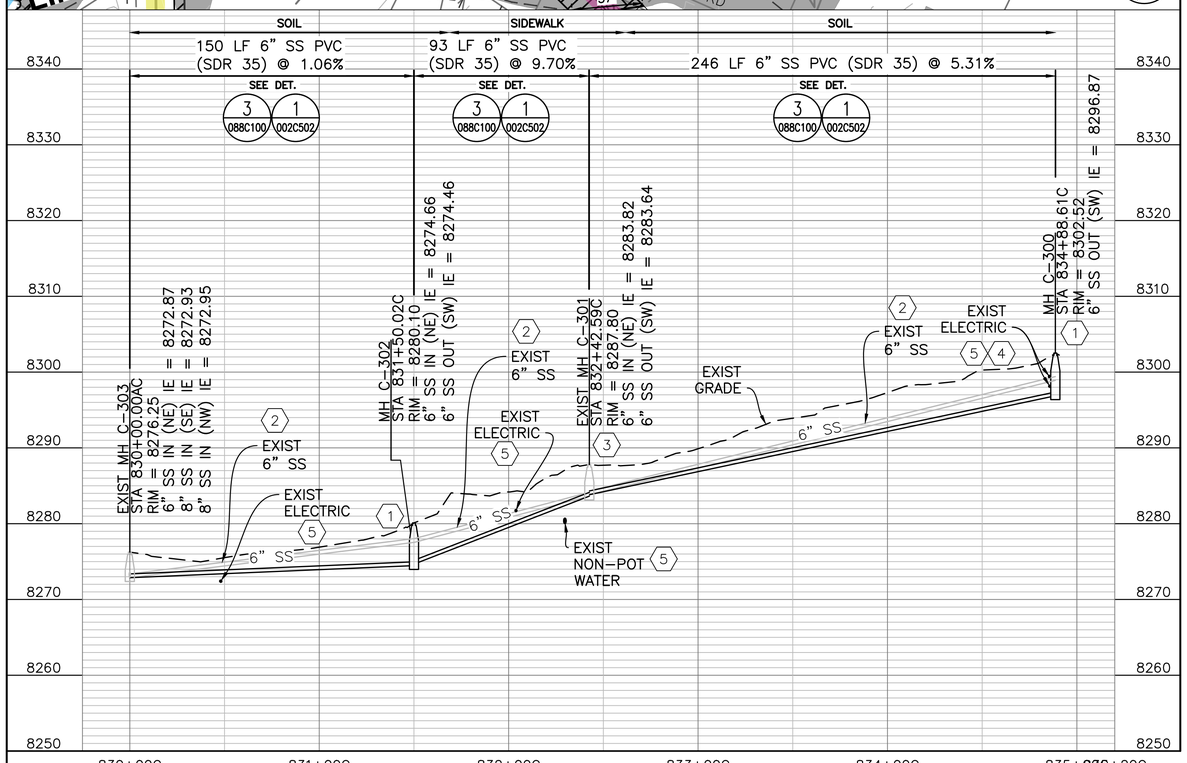
1" = 80'

Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/21/25 ARIZONA, U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	086 C100	UTILITIES – CONCESSIONAIRE AREA (REGION 3) SANITARY SEWER PLAN KEY MAP	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	CADD: SJC			PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 32 OF 36
	DATE: AUG 2026		NORTH RIM – GRAND CANYON NATIONAL PARK	





LINE C

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
4. DRY UTILITY TO BE REROUTED AROUND STRUCTURE
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

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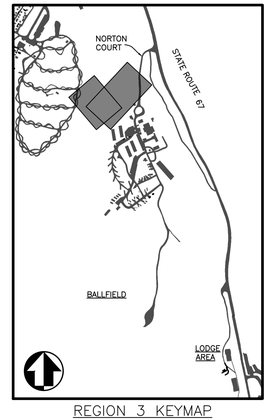
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2–INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

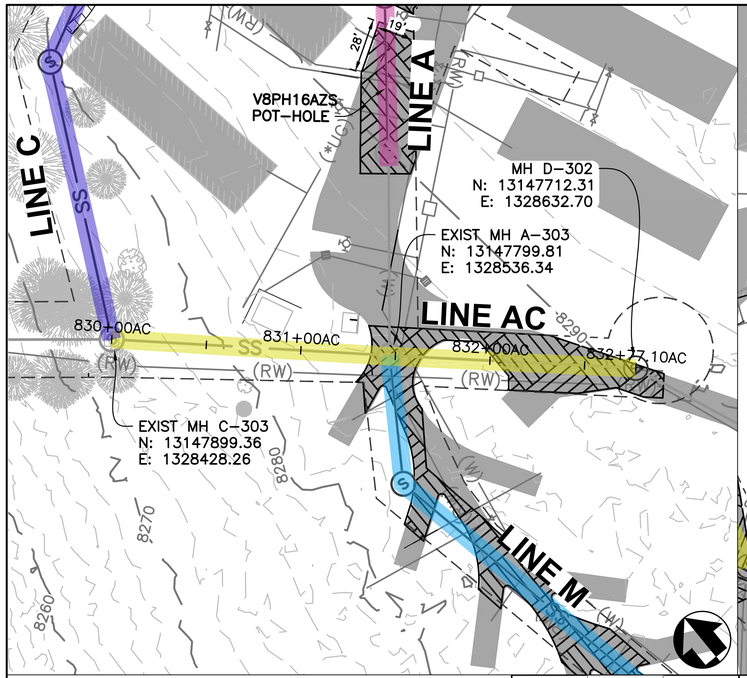


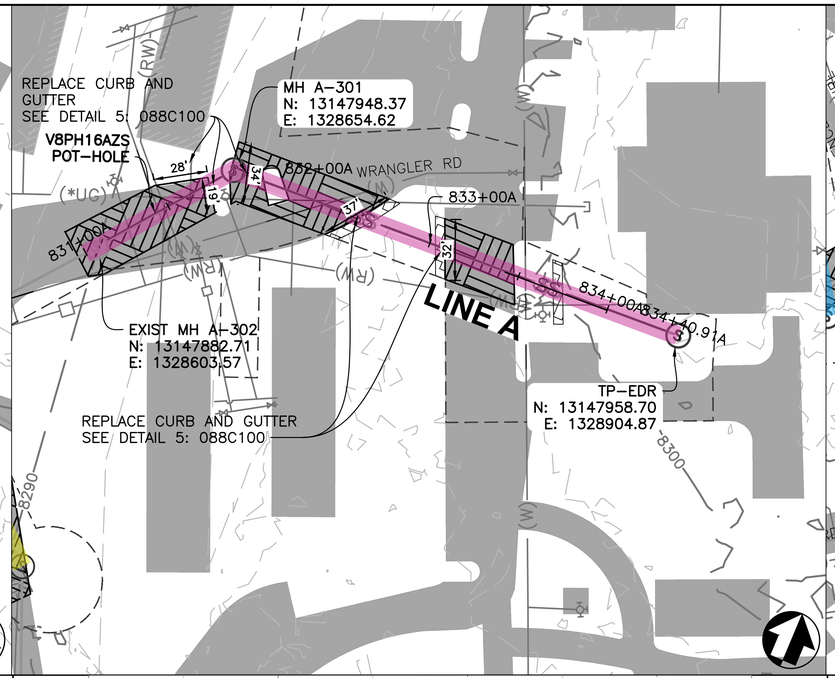
REGION 3 KEYMAP

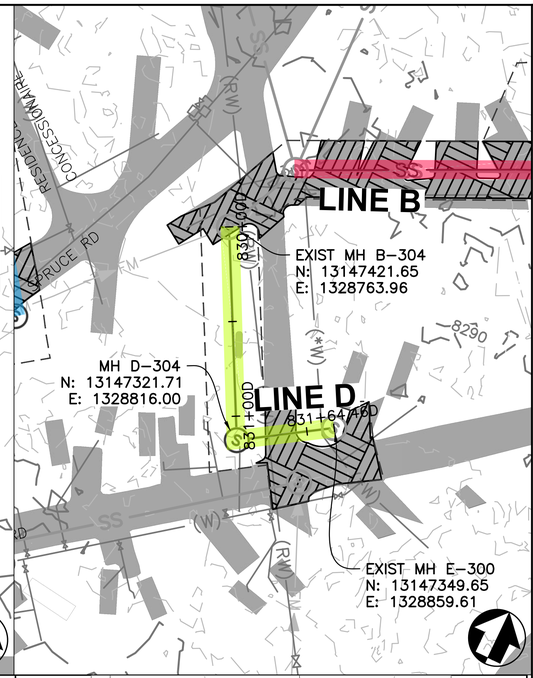
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

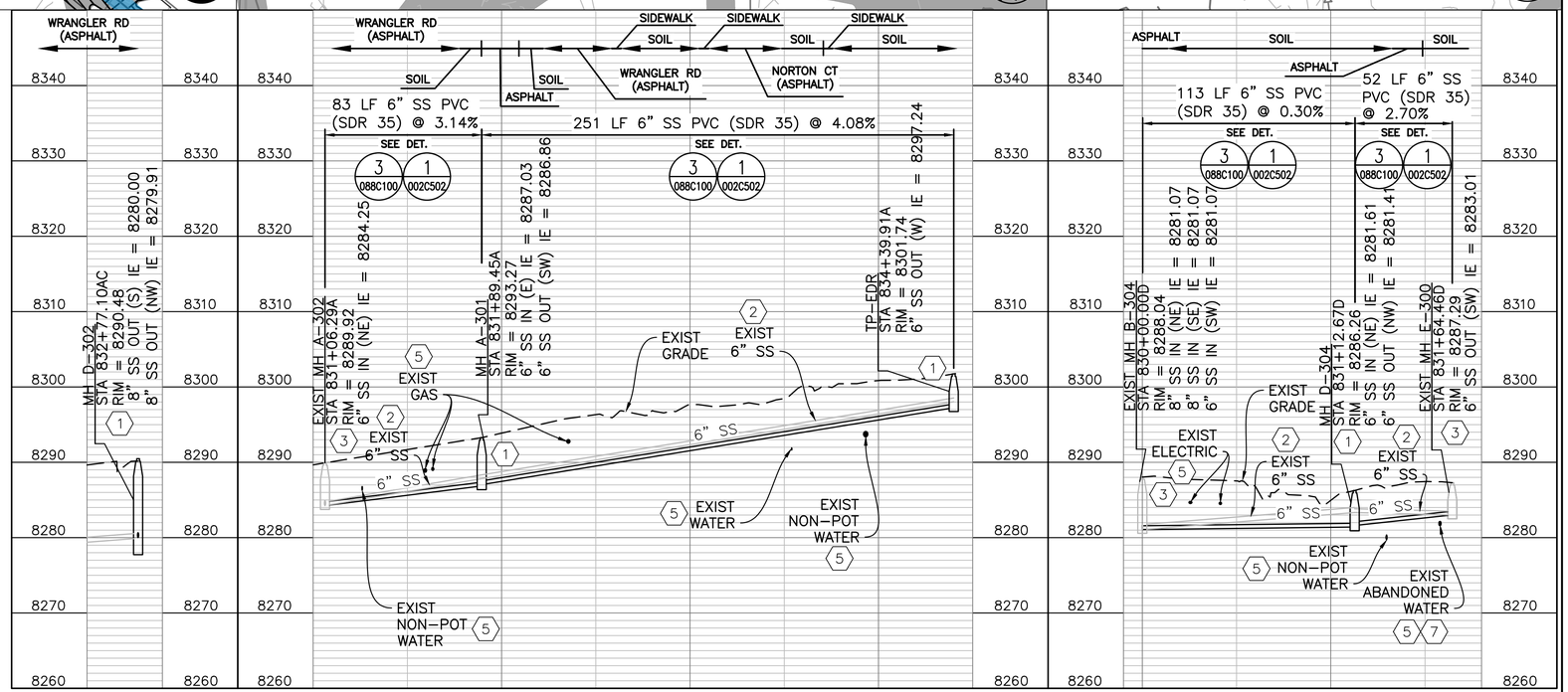
Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS–HOTOPP Date Signed 9/21/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 086 C101	TITLE OF SHEET UTILITIES – CONCESSIONAIRE AREA (REGION 3) SANITARY SEWER PLAN & PROFILE LINE C NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 33 OF 36









LINE AC LINE A LINE D

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
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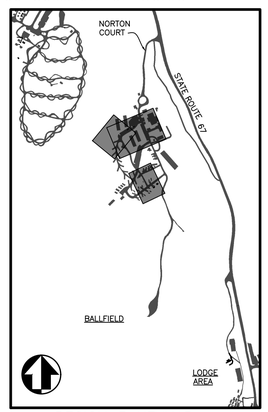


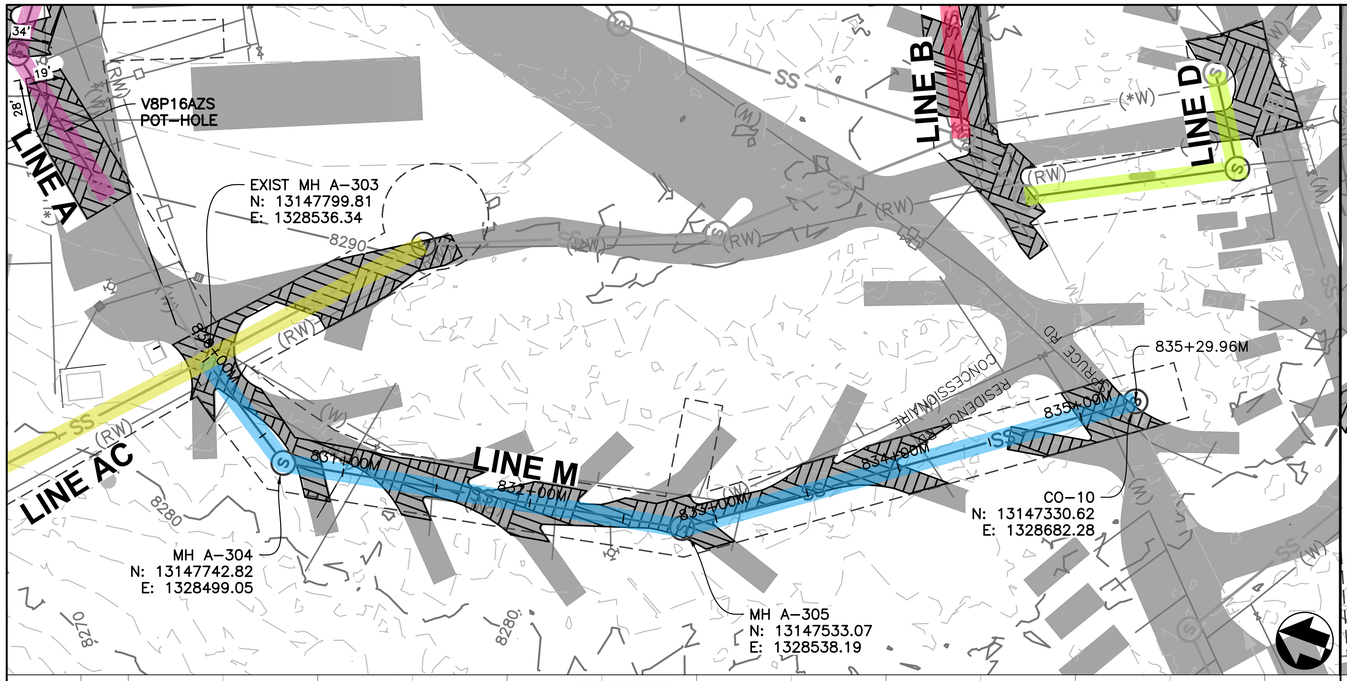
REGION 3 KEYMAP

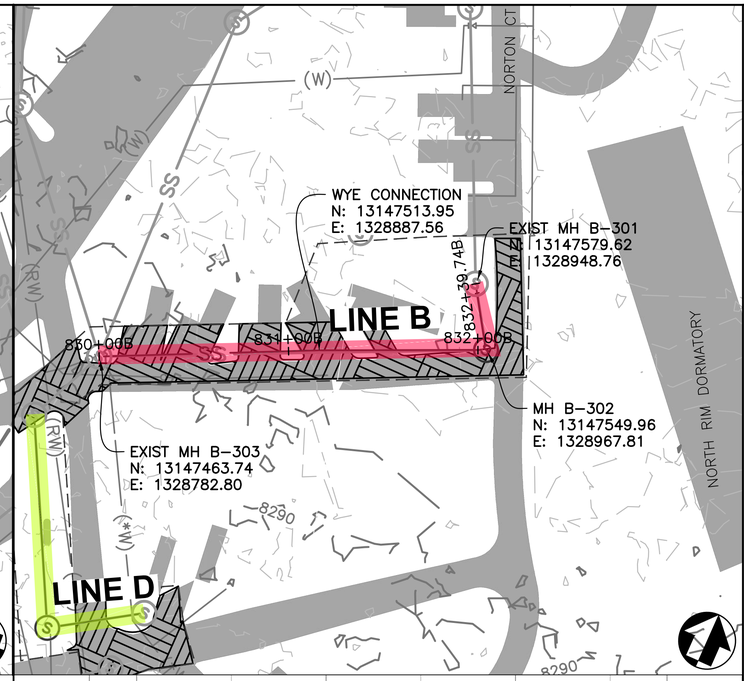
40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

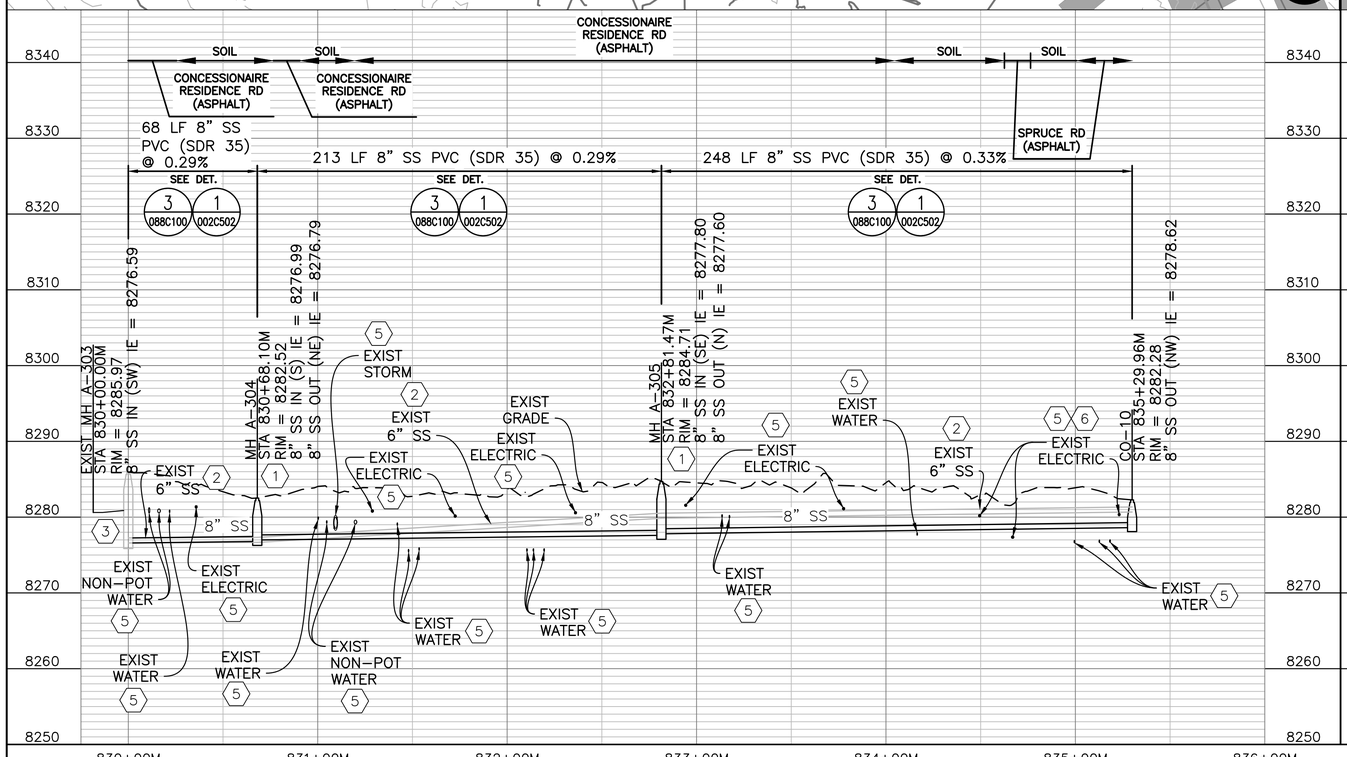
Registered Professional Engineer (Civil) CERTIFICATE NO. 77465 BRIAN STEPHENS-HOTOPP Date Signed 9/21/25 ARIZONA, U.S.A.

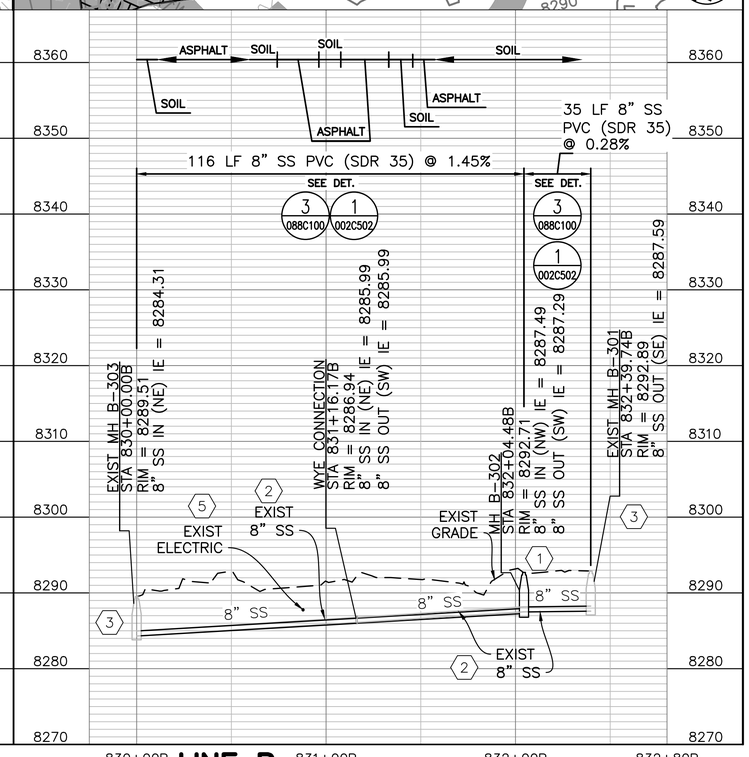
Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	086 C102	UTILITIES – CONCESSIONAIRE AREA (REGION 3) SANITARY SEWER PLAN & PROFILE LINES AC, A, D NORTH RIM – GRAND CANYON NATIONAL PARK	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DADE SJC			PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 34 OF 36
	DATE: AUG 2026			









LINE M

LINE B

KEY NOTES

1. DEMO & REPLACE EXISTING STRUCTURE
2. DEMO & REMOVE EXISTING PIPE
3. CONNECT TO EXISTING STRUCTURE
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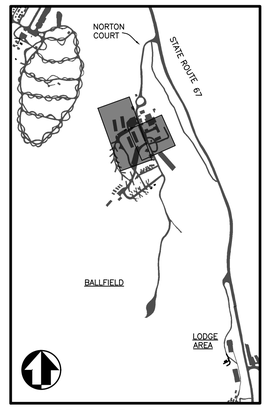
5. FIELD VERIFY EXISTING UTILITY LOCATIONS
6. RAISE OR LOWER DRY UTILITY TO PROVIDE MINIMUM 12" CLEARANCE TO SANITARY PIPE
7. FIELD VERIFY LINE IS ABANDONED. PORTION OF ABANDONED LINE IN CONFLICT WITH PROJECT CONSTRUCTION TO BE REMOVED 5' IN EACH DIRECTION.

GENERAL NOTES:

A. ALL EXISTING SERVICES TO BE RECONNECTED TO NEW PIPE.
B. SURFACE FEATURES TO BE RETURNED TO ORIGINAL CONDITION OR IN KIND FOLLOWING CONSTRUCTION, IN COORDINATION WITH THE CONTRACTING OFFICER. REFER TO VOLUME 0, SHEET 002C506 FOR PAVING DETAILS.
C. ALL UTILITY LOCATIONS ARE APPROXIMATE. CONTRACTOR IS RESPONSIBLE FOR VERIFYING ALL UTILITY CROSSINGS AND MINIMUM CLEARANCES AS SPECIFIED IN THE KEYNOTES.
D. IN AREAS WHERE PIPE COVER IS LESS THAN 42 INCHES BELOW GRADE, PROVIDE 2-INCH THICK BLUEBOARD (XPS) PLACED 6 TO 8 INCHES ABOVE THE PIPE AND EXTENDED TO 2 FEET ON EITHER SIDE OF THE PIPE OUTSIDE DIAMETER.
E. SEE DETAIL 1/002C504 FOR WATER/SEWER CROSSING REQUIREMENTS, INCLUDING REQUIRED PIPE CLEARANCE AND MATERIAL.
F. DRAWINGS IN THIS VOLUME REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.



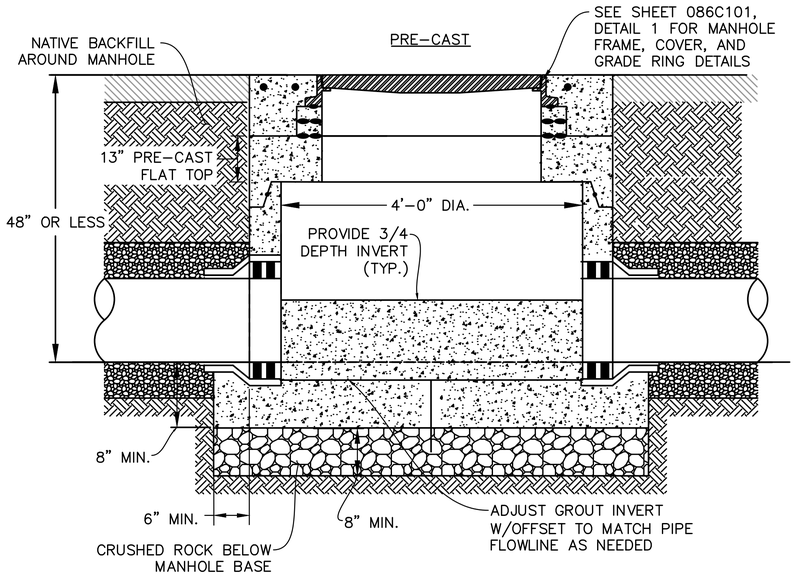
REGION 3 KEYMAP

40 0 40 80 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS-HOTOPP Date Signed 9/21/26 ARIZONA, U.S.A.

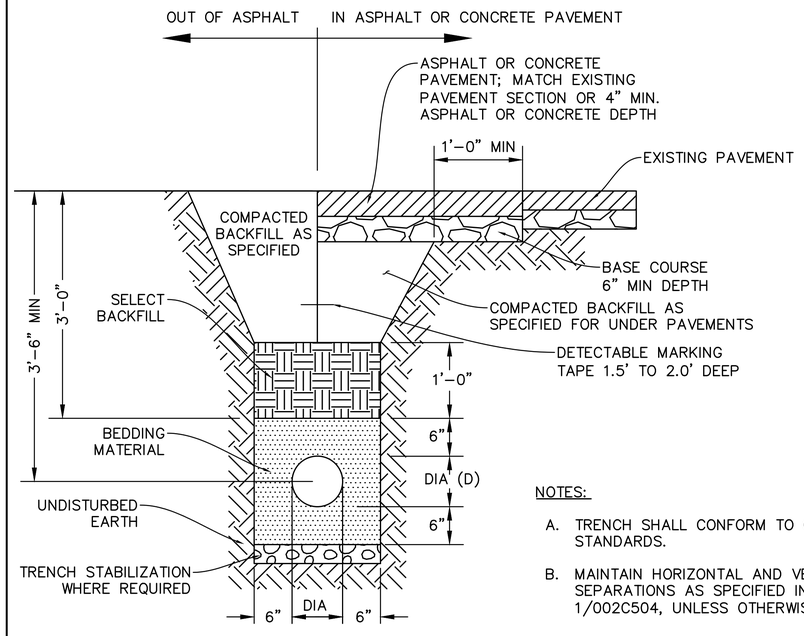
Brian Stephens-Hotopp

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	DESIGNED: KNF/SJC DRAWN: SJC TECH. REVIEW: BASH DATE: AUG 2026	SUB SHEET NO. 086 C103	TITLE OF SHEET UTILITIES – CONCESSIONAIRE AREA (REGION 3) SANITARY SEWER PLAN & PROFILE LINES M, B NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 35 OF 36



1 088C100 STANDARD SHALLOW SANITARY SEWER MANHOLE

SCALE NTS



NOTES:

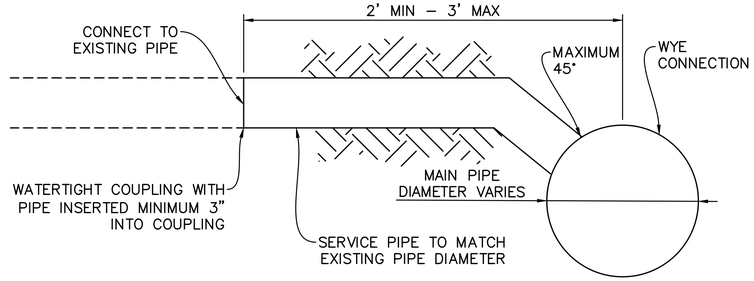
A. TRENCH SHALL CONFORM TO OSHA STANDARDS.

B. MAINTAIN HORIZONTAL AND VERTICAL SEPARATIONS AS SPECIFIED IN DETAIL 1/002C504, UNLESS OTHERWISE NOTED.

PIPE TRENCH

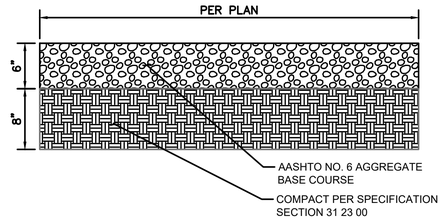
3 088C100 PIPE TRENCH

SCALE NTS



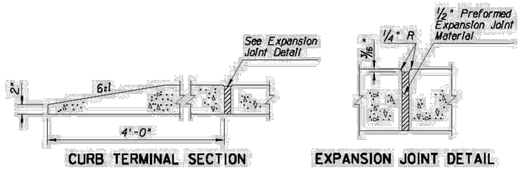
2 088C100 STANDARD SANITARY SEWER SERVICE CONNECTION

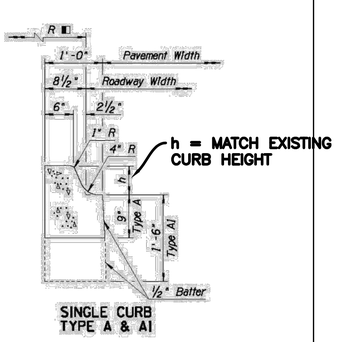
SCALE NTS



4 088C100 GRAVEL PATHWAY DETAIL

SCALE NTS





GENERAL NOTES

SINGLE CURB AND CURB & GUTTER

1. Single curb and curb & gutter may be constructed by the use of forms or the concrete may be extruded.
2. When the pavement section slopes away from the gutter, the slope of the gutter shall match the pavement cross slope. Therefore, the gutter depression is not applicable.
3. Two-inch deep contraction joints shall be placed in the curb and the gutter at locations which match the joints in adjacent PCCP and at approximate 15' centers when adjacent to AC pavement. Joints shall be either hand-tooled or sawn.
4. Expansion joints shall be located at tangent points in curb returns, at structures and at maximum 60' intervals. The ½" joint filler shall extend the full depth of the concrete.
5. Concrete shall be finished with a steel trowel followed by brushing with a fine brush along the length of the curb and gutter.
6. All exposed edges and hand-tooled joints shall be finished with a tool having a ¼" radius, or as noted on the plans.
7. Place AB under single curb, valley gutter, and curb & gutter when shown on plans

APPROVED FOR DESIGN	STATE OF ARIZONA DEPARTMENT OF TRANSPORTATION ROADWAY STANDARD DRAWINGS	REV. 5/12
APPROVED FOR DISTRIBUTION	CURB & GUTTER CURB GUTTER ③	DRAWING NO. C-05.10

5 088C100 CURB AND GUTTER DETAIL

SCALE NTS

Registered Professional Engineer (Civil) CERTIFICATE No. 77465 BRIAN STEPHENS-HOTOPP Date Signed 8/14/26 ARIZONA U.S.A.

Brian Stephens-Hotopp

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	KNF/SJC	088 C100	UTILTIIES SANITARY SEWER DETAILS	113 182752
SUBCONTRACTOR: CALIBRE ENGINEERING, INC HIGHLANDS RANCH, CO	ADD SJC			PMIS/PKG NO. 318719
	TECH. REVIEW: BASH			SHEET 36 OF 36
	DATE: AUG 2026		NORTH RIM – GRAND CANYON NATIONAL PARK	